

THE COHOMOLOGICAL HALL ALGEBRA OF THE RESOLVED CONIFOLD: MASTER SYNTHESIS

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ABSTRACT. The critical cohomological Hall algebra of the resolved conifold $\mathbf{Y} = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1))$ admits five independent constructions — direct Klebanov–Witten super-shuffle, two-chart Čech gluing of two $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks, Van den Bergh tilting transport from $D^b(\text{Coh } \mathbf{Y})$ to $D^b(J(Q_{\text{con}}, W_{\text{con}})\text{-mod})$, RTT super-Yangian via Gauss decomposition, and chiral vertex-algebra realisation via the Gaiotto–Rapčák corner $Y_{0,1,1}[\psi]$ under the Li–Yamazaki toric-quiver dictionary (R5 is conjectural at non-generic ψ , proved on the self-dual locus $\epsilon_1 + \epsilon_2 = 0$; see Remark 8.1). All five agree on the leading structure constant at the $(1, 1)$ -weight imaginary root; the subleading c_{11} asymmetric correction is conjectural (Conjecture 9.10, Remark 9.11). The Drinfeld double is the affine super-Yangian $Y(\widehat{\mathfrak{gl}}(1|1))$ at rational, trigonometric, and toroidal levels; the elliptic lift is quasi-Hopf via the Felder–Jimbo–Konno dynamical R -matrix. Explicit status markers distinguish theorems, propositions, conjectures, convention-dependent statements, and open items. The techniques transfer to toric CY_3 quivers; a sharp obstruction prevents direct transport to compact CY_3 , where automorphic-form machinery (Borcherds products, Gritsenko–Nikulin) replaces chart-wise shuffle presentations.

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1. GEOMETRIC DATUM

The resolved conifold

$$\mathbf{Y} = \text{Tot}(\pi: E \rightarrow \mathbb{P}^1), \quad E = \mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1),$$

is a smooth quasi-projective Calabi–Yau threefold of complex dimension three. As a rank-2 holomorphic vector bundle over $\mathbb{P}^1$ its topological Euler number is $\chi_{\text{top}}(\mathbf{Y}) = \chi_{\text{top}}(\mathbb{P}^1) = 2$; its holomorphic Euler characteristic $\chi(\mathcal{O}_{\mathbf{Y}})$ is infinite (global sections of $\mathcal{O}_{\mathbf{Y}}$ include all polynomials in the fibre coordinates). The second homology $H_2(\mathbf{Y}; \mathbb{Z}) = \mathbb{Z}[\mathbb{P}^1]$ is generated by the class of the zero section, a rational curve of genus zero; the effective cone of curve classes is $\mathbb{Z}_{\geq 0}\beta$ with $\beta = [\mathbb{P}^1]$.

Triviality of the canonical bundle is a first-principles consequence of the total-space formula: for any vector bundle E over a smooth base B the canonical bundle of $\text{Tot}(E)$ is $K_{\text{Tot}(E)} = \pi^*(K_B \otimes \det(E^\vee))$ (Hartshorne III. 8, relative-cotangent sequence). Here $K_{\mathbb{P}^1} = \mathcal{O}(-2)$ and $\det(E^\vee) = \mathcal{O}(1) \oplus \mathcal{O}(1)$ -determinant $= \mathcal{O}(2)$, so $K_{\mathbf{Y}} = \pi^* \mathcal{O}_{\mathbb{P}^1} = \mathcal{O}_{\mathbf{Y}}$. A nowhere-vanishing global $(3, 0)$ -form $\Omega_{\mathbf{Y}}$ is then determined up to $\mathbb{C}^\times$; in the affine chart U_+ below it reads $\Omega_{\mathbf{Y}}|_{U_+} = dz \wedge du_+ \wedge dv_+$, and the Jacobian of the transition (1) gives $\Omega_{\mathbf{Y}}|_{U_-} = -d\tilde{z} \wedge du_- \wedge dv_-$ on U_- (sign tracked in Section 4).

First Dolbeault vanishing $H^{0,1}(\mathbf{Y}) = H^1(\mathbf{Y}, \mathcal{O}_{\mathbf{Y}}) = 0$ follows from the Leray spectral sequence for π : since E is a direct sum of negative line bundles, $\pi_* \mathcal{O}_{\mathbf{Y}} = \text{Sym}^\bullet(E^\vee) = \bigoplus_{q \geq 0} \text{Sym}^q(\mathcal{O}(1) \oplus \mathcal{O}(1))$ is a direct sum of line bundles of non-negative degree on $\mathbb{P}^1$, and higher direct images $R^{\geq 1} \pi_* \mathcal{O}_{\mathbf{Y}} = 0$ because π is affine. Hence $H^1(\mathbf{Y}, \mathcal{O}_{\mathbf{Y}}) = H^1(\mathbb{P}^1, \pi_* \mathcal{O}_{\mathbf{Y}}) = \bigoplus_{q \geq 0} H^1(\mathbb{P}^1, \text{Sym}^q(\mathcal{O}(1) \oplus \mathcal{O}(1))) = 0$, all summands being H^1 of a non-negative line bundle on $\mathbb{P}^1$.

A two-chart atlas $\mathbf{Y} = U_+ \cup U_-$ with $U_\pm \cong \mathbb{C}^3$ arises from the standard affine cover of $\mathbb{P}^1$; the overlap is $U_+ \cap U_- \cong \mathbb{C}^\times \times \mathbb{C}^2$. In coordinates (z, u_+, v_+) on U_+ and $(\tilde{z}, u_-, v_-)$ on U_- the transition reads

$$(1) \quad \tilde{z} = z^{-1}, \quad u_- = zu_+, \quad v_- = zv_+,$$

encoding the $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ twist. As a complex variety each chart $U_\pm \cong \mathbb{C}^3$ carries the Schiffmann–Vasserot critical CoHA $Y^+(\widehat{\mathfrak{gl}}_1)$ associated to the Hilbert scheme $\text{Hilb}^\bullet(\mathbb{C}^3)$, presented in the Kontsevich–Soibelman dimensional-reduction framework by the one-node three-loop quiver with abstract loop generators X, Y, Z and cubic potential $W_{\text{Jor}} = \text{tr } X[Y, Z]$ (Schiffmann–Vasserot [arXiv:1202.2756](#) Thm. 1.1). The quiver variables X, Y, Z are module-level endomorphisms at a single node; they are not the chart coordinates $(z, u_\pm, v_\pm)$ of $U_\pm$. The identification $\text{CoHA}_\pm(U_\pm) = Y^+(\widehat{\mathfrak{gl}}_1)$ is the critical-locus equivalence $\text{Crit}(W_{\text{Jor}})_n = \{(X, Y, Z) \in \mathfrak{gl}_n^3 : [X, Y] = [Y, Z] = [Z, X] = 0\} // \text{GL}_n$ with cyclic stability and vanishing-cycle perverse sheaf, matching $\text{Hilb}^n(\mathbb{C}^3)$ at dimension vector n .

The toric structure of $\mathbf{Y}$ is a three-dimensional algebraic torus $T^3 = (\mathbb{C}^\times)^3$ with weight data $(a, b, c) \in \mathbb{Z}^3$ on (z, u, v) : z carries the $\mathbb{P}^1$ -rotation weight a , $u_\pm$ and $v_\pm$ carry independent fibre weights b and c . The T^3 -fixed locus is the pair of toric points $p_\pm = \{z = 0 \text{ or } \infty\} \cap \{u = v = 0\} \subset \mathbf{Y}$, one per chart. The Calabi–Yau subtorus $T_{\text{CY}}^2 \subset T^3$ preserving $\Omega_{\mathbf{Y}}$ is cut out by the single linear relation $a + b + c = 0$ on weights (Bridgeland [math/0502050](#) §3; Klebanov–Witten [hep-th/9807080](#) §2). Equivariant cohomology parameters $(\varepsilon_1, \varepsilon_2, \varepsilon_3) \in H_{T^3}^2(\text{pt}) = \text{Sym}^1(\text{Lie}(T^3)^*)$, dual to the weight coordinates (a, b, c) , inherit the CY constraint $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ upon restriction to $H_{T_{\text{CY}}^2}^*$. The refined Neguț kernel $(u + \varepsilon_1)(u + \varepsilon_2)/[u(u + \varepsilon_1 + \varepsilon_2)]$ is a T^3 -equivariant rational function on the shuffle plane (Section 4); its $\mathbb{C}^3$ -vertex limit to the Jordan-quiver kernel $(u + \varepsilon_1)(u + \varepsilon_2)(u + \varepsilon_3)/u^3$

and its CY-specialisation to $\varepsilon_1 + \varepsilon_2 + \varepsilon_3 = 0$ are orthogonal specialisations carried out elsewhere in this synthesis.

2. THE KLEBANOV–WITTEN NON-COMMUTATIVE CREPANT RESOLUTION

Theorem 2.1 (Van den Bergh 2004, [arXiv:math/0211064](#) Prop. 3.2.10). **[Proved elsewhere]** *The tilting bundle $T = \mathcal{O}_{\mathbf{Y}} \oplus \mathcal{O}_{\mathbf{Y}}(1)$ on the resolved conifold $\mathbf{Y}$ satisfies $\text{End}_{\mathbf{Y}}(T) \cong J(Q_{\text{con}}, W_{\text{con}})$, the Jacobi algebra of the Klebanov–Witten quiver*

$$Q_{\text{con}} : \quad 0 \xrightleftharpoons[b_1, b_2]{a_1, a_2} 1$$

with quartic potential

$$W_{\text{con}} = \text{tr}(a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1).$$

The derived Morita equivalence is the adjoint pair

$$L = (-) \otimes_{J(Q_{\text{con}}, W_{\text{con}})}^L T : D^b(J(Q_{\text{con}}, W_{\text{con}})\text{-mod}) \rightleftarrows D^b(\text{Coh } \mathbf{Y}) : R = R\text{Hom}_{\mathbf{Y}}(T, -),$$

with unit $\eta : \text{id} \Rightarrow RL$ and counit $\varepsilon : LR \Rightarrow \text{id}$ natural isomorphisms. Both directions rest on two load-bearing facts: (i) T compactly generates $D^b(\text{Coh } \mathbf{Y})$ (Bondal–Van den Bergh [arXiv:math/0204218](#) Thm. 3.1.1, since $T|_{\mathbb{P}^1} = \mathcal{O} \oplus \mathcal{O}(1)$ is Beilinson’s generator of $D^b(\mathbb{P}^1)$ and $\pi : \mathbf{Y} \rightarrow \mathbb{P}^1$ preserves compact generation); (ii) $\text{Ext}_{\mathbf{Y}}^0(T, T) = 0$, reducing to $H^{\geq 1}(\mathbb{P}^1, \mathcal{O}(-1)) = 0$ via the Leray spectral sequence on $\mathbf{Y} = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1))$. The Calabi–Yau–3 structure lives on the Ginzburg dg-lift $\Gamma_{\text{con}} := \Gamma(Q_{\text{con}}, W_{\text{con}})$, a dg-algebra concentrated in degrees $[-2, 0]$ with $H^0(\Gamma_{\text{con}}) = J(Q_{\text{con}}, W_{\text{con}})$ and bimodule self-duality

$$\Gamma_{\text{con}} \xrightarrow{\sim} R\text{Hom}_{\Gamma_{\text{con}} \otimes \Gamma_{\text{con}}^{\text{op}}}(\Gamma_{\text{con}}, \Gamma_{\text{con}} \otimes \Gamma_{\text{con}})[3]$$

(Ginzburg [arXiv:math/0612139](#) Thm. 3.2.8; Keller–Van den Bergh [arXiv:0906.0761](#) Thm. 6.3 for the deformed Calabi–Yau completion). Bocklandt [arXiv:math/0603558](#) Thm. 3.1 verifies non-degeneracy: each of a_1, a_2, b_1, b_2 appears in W_{con} ; the four Jacobi relations (2) are linearly independent; the cyclic pairing on $HC_{\bullet}(\Gamma_{\text{con}})$ is non-degenerate. Consequently Γ_{con} is quasi-isomorphic to $J(Q_{\text{con}}, W_{\text{con}})$ concentrated in degree zero, the Serre functor on $\text{Perf}(\Gamma_{\text{con}})$ is the shift $[3]$, and $D^b(J(Q_{\text{con}}, W_{\text{con}})\text{-mod})$ inherits the CY-3 structure.

The four cyclic derivatives of W_{con} are computed directly from the cyclic-derivative rule $\partial_x W = \sum_{W=uxv} vu$:

$$(2) \quad \begin{aligned} \partial_{a_1} W &= b_1 a_2 b_2 - b_2 a_2 b_1, & \partial_{b_1} W &= a_2 b_2 a_1 - a_1 b_2 a_2, \\ \partial_{a_2} W &= b_2 a_1 b_1 - b_1 a_1 b_2, & \partial_{b_2} W &= a_1 b_1 a_2 - a_2 b_1 a_1. \end{aligned}$$

Each relation (2) equates two length-3 monomials built from a fixed pair of arrow-types; the difference is the image under the A_{∞} -transfer of a single Maurer–Cartan bracket $[\mu_2, \mu_2]$ on the tensor coalgebra, and encodes one quartic cyclic derivative of W_{con} . The conifold coordinate ring is the centre of the Jacobi algebra, not its abelianisation: by Van den Bergh [arXiv:math/0211064](#) Prop. 3.2.10, elements of $Z(J(Q_{\text{con}}, W_{\text{con}}))$ are idempotent-diagonal pairs

$$x = a_1 b_1 \oplus b_1 a_1, \quad y = a_2 b_2 \oplus b_2 a_2, \quad z = a_1 b_2 \oplus b_2 a_1, \quad w = a_2 b_1 \oplus b_1 a_2,$$

in $e_0 J e_0 \oplus e_1 J e_1$. The Jacobi relations (2) imply $xy = zw$ in the centre, giving the ring isomorphism

$$Z(J(Q_{\text{con}}, W_{\text{con}})) \cong \mathbb{C}[x, y, z, w]/(xy - zw),$$

the coordinate ring of the singular affine conifold X_0 . The NCCR morphism $\text{Spec } J(Q_{\text{con}}, W_{\text{con}}) \rightarrow \text{Spec } Z(J(Q_{\text{con}}, W_{\text{con}})) = X_0$ is Van den Bergh’s non-commutative resolution. The abelianisation $J/[J, J]$ is a distinct (larger) module: in $J/[J, J]$ every commutator $b_1 a_2 b_2 - b_2 a_2 b_1$ vanishes tautologically, so the abelianisation cuts out no variety beyond the (e_0, e_1) vertex-grading.

The two simple modules S_0, S_1 of $J(Q_{\text{con}}, W_{\text{con}})$ at vertices 0, 1 have Ext-dimensions

(a, b)	$\text{Ext}^0(S_a, S_b)$	Ext^1	Ext^2	Ext^3	$\chi(S_a, S_b)$	$P(S_a, S_b)$
$(0, 0)$	1	0	0	1	0	2
$(0, 1)$	0	2	2	0	0	4
$(1, 0)$	0	2	2	0	0	4
$(1, 1)$	1	0	0	1	0	2

by CY-3 Serre duality $\text{Ext}^i(S_a, S_b) \cong \text{Ext}^{3-i}(S_b, S_a)^\vee$, with $\text{Ext}^1(S_0, S_1) = \#\{a_1, a_2\} = 2$ and $\text{Ext}^1(S_1, S_0) = \#\{b_1, b_2\} = 2$. Here $\chi(S_a, S_b) = \sum_i (-1)^i \dim \text{Ext}^i(S_a, S_b)$ is the Euler form and $P(S_a, S_b) = \sum_i \dim \text{Ext}^i(S_a, S_b)$ is the total Poincaré dimension; the two are distinct invariants.

Remark 2.2 (The Euler form is antisymmetric on CY-3). **[Convention-dependent]** On a Calabi–Yau d -category, Serre duality gives $\text{Ext}^i(\gamma, \gamma') \cong \text{Ext}^{d-i}(\gamma', \gamma)^\vee$, so

$$\chi(\gamma, \gamma') = \sum_i (-1)^i \dim \text{Ext}^i(\gamma, \gamma') = (-1)^d \chi(\gamma', \gamma).$$

For $d = 2$ the Euler form is symmetric (K3 Mukai pairing); for $d = 3$ the Euler form is antisymmetric, in particular $\chi(\gamma, \gamma) = 0$. The vanishing $\chi(S_a, S_a) = 1 - 0 + 0 - 1 = 0$ in the table above is the CY-3 antisymmetry on the diagonal, not a separate “skew-symmetrisation.” The Kontsevich–Soibelman quantum-torus bracket $[y^\alpha, y^\beta] = (-\mathbb{L}^{1/2})^{\chi(\alpha, \beta)} (1 - \mathbb{L}^{\chi(\alpha, \beta)}) y^{\alpha+\beta} + \dots$ uses this antisymmetric χ directly. The total Poincaré dimension $P(S_a, S_a) = 2 = \dim \text{Ext}^0 + \dim \text{Ext}^3$ counts simples-plus-duals and is distinct from χ ; the two agree only on CY-even categories or when $\text{Ext}^{\text{odd}} = 0$.

Remark 2.3 (NCCR is not McKay: the conifold is not an orbifold). **[Proved elsewhere]** The singular conifold $X_0 = \text{Spec } \mathbb{C}[x, y, z, w]/(xy - zw)$ is not a quotient $\mathbb{C}^3/G$ for any finite $G \subset \text{SL}_3(\mathbb{C})$: the class group $\text{Cl}(X_0) = \mathbb{Z}$ is infinite, while any Gorenstein quotient singularity has finite class group of order dividing $|G|$. The derived McKay correspondence of Bridgeland–King–Reid [arXiv:math/9908027](#), which gives $D^b(\text{Coh } \mathbb{C}^3/G) \simeq D^b(\text{Coh } \text{Hilb}^G \mathbb{C}^3)$ for $G \subset \text{SL}_3(\mathbb{C})$ finite, does not apply to the conifold; the Klebanov–Witten NCCR is the genuine non-orbifold replacement (Szendrői [arXiv:0705.3419](#)). The quartic degree of W_{con} reflects this: the local- $\mathbb{P}^2$ McKay-orbifold $\mathbb{C}^3/\mathbb{Z}_3$ carries the Beilinson 9-arrow quiver with *cubic* potential $W_{\mathbb{P}^2} = \text{tr}(x[y, z])$ -style (cycle through compact divisor), while the conifold has no compact divisor and its quiver cycles are length four through the two $\mathbb{P}^1$ -nodes. Potential degree is a brane-tiling invariant (Hanany–Kennaway [hep-th/0503149](#); Franco–Hanany–Kennaway–Vegh–Wecht [hep-th/0511063](#)), not a direct cohomological number.

3. THE SUPER ROOT SYSTEM AND BPS LIE ALGEBRA

Theorem 3.1 (Morrison–Mozgovoy–Nagao–Szendrői, [arXiv:1107.5017](#) Thm. 1). **[Proved elsewhere]** *The motivic Donaldson–Thomas partition function for $\mathbf{Y}$ in the non-commutative (Szendrői) chamber is*

$$Z_\zeta^{\text{NC}}(y_0, y_1) = \prod_{\alpha \in \Delta_+^{\text{re}}} \prod_{j \geq 0} (1 - \mathbb{L}^{-1/2+j} y^\alpha)^{-1} \cdot \prod_{\alpha \in \Delta_+^{\text{im}}} \prod_{j=0}^1 (1 - \mathbb{L}^{-1/2+j} y^\alpha)^{-1}$$

with root decomposition

$$\Delta_+^{\text{re}} = \{(i+1, i), (i, i+1) : i \geq 0\}, \quad \Delta_+^{\text{im}} = \{(n, n) : n \geq 1\},$$

the affine $A_1^{(1)}$ positive root system. Motivic BPS indices are $\Omega^{\text{mot}}(i+1, i) = \Omega^{\text{mot}}(i, i+1) = 1$ (a single odd generator per real root) and $\Omega^{\text{mot}}(n, n) = -\mathbb{L} - 1$, specialising at $\mathbb{L} = 1$ to the numerical $\Omega^{\text{num}}(n, n) = -2$. The sign is the Kontsevich–Soibelman parity phase for a bosonic-centre imaginary-root direction (Mozgovoy–Reineke [arXiv:0708.1259](#) Prop. 5.2); the unsigned super-dimension of the imaginary-root space of $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ at (n, n) is 2, a two-dimensional span(H_n, k_n) carried in Remark 3.3.

The super-PBW decomposition of Davison–Hennecart–Schlegel-Mejia [arXiv:2303.12592](#) (extending the ungraded Davison–Meinhardt PBW [arXiv:1601.02479](#) to $\mathbb{Z}/2$ -graded CoHA) gives, as graded super vector spaces,

$$\text{CoHA}(Q_{\text{con}}, W_{\text{con}}) \cong \text{Sym}_{\text{super}}^{\boxplus+}(H(BC^\times)_{\text{vir}} \otimes \mathfrak{g}_{\text{BPS}}^{\text{con}}),$$

with $H(BC^\times)_{\text{vir}} = \mathbb{Q}[u]$ the polynomial ring on one generator in cohomological degree 2, $\text{Sym}_{\text{super}}^{\boxplus+}$ the super-symmetric (Koszul-signed) product, and $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ the *super BPS Lie algebra*: the primitive part under the Hopf coproduct, supported on $\Delta_+^{\text{re}} \cup \Delta_+^{\text{im}}$ with super-dimensions matching the MMNS multiplicities. The Mozgovoy–Reineke optimality of the Klebanov–Witten potential ([arXiv:0708.1259](#) Prop. 5.2, equivalently the Ginzburg dga vanishing $\text{HH}^{<0}(J) = 0$) is the technical input that validates the super-PBW transport.

Proposition 3.2 (Davison–Meinhardt integrality for $\widehat{\mathfrak{gl}}(1|1)$). **[Proposition]** *The Davison–Meinhardt integrality theorem ([arXiv:1601.02479](#) Thm. A) asserts that the motivic BPS invariant $\Omega^{\text{mot}}(\alpha) \in \mathbb{Z}[\mathbb{L}^{1/2}, \mathbb{L}^{-1/2}]$ is a polynomial with integer coefficients when expressed in the supersymmetric plethystic variable, equivalently that the PBW factorisation of the motivic DT partition function lifts to a $\mathbb{Z}$ -integral super-vector space on each root direction. Specialising to $(Q_{\text{con}}, W_{\text{con}})$ with MMNS multiplicities $(\Omega^{\text{mot}}(\text{re}), \Omega^{\text{mot}}(\text{im})) = (1, -\mathbb{L} - 1)$:*

- (i) *On the real roots $\alpha \in \Delta_+^{\text{re}}$, integrality gives a single odd super-generator (parity from the KS phase of a fermionic root direction), contributing an exterior tensor factor $\Lambda^\bullet(\mathbb{Q}[u] \otimes \mathfrak{g}_{\text{BPS}, \alpha}^{\text{con}})$ in the super-symmetric algebra with super-dimension $\dim_{\text{super}} \mathfrak{g}_{\text{BPS}, \alpha}^{\text{con}} = 0|1$ (zero even, one odd).*
- (ii) *On the imaginary roots $(n, n) \in \Delta_+^{\text{im}}$, the numerical limit $\mathbb{L} \rightarrow 1$ of $\Omega^{\text{mot}}(n, n) = -\mathbb{L} - 1$ yields $\Omega^{\text{num}}(n, n) = -2$; the minus sign is the Kontsevich–Soibelman parity phase (Mozgovoy–Reineke [arXiv:0708.1259](#) Prop. 5.2) encoding that the imaginary-root contribution is a 2-dimensional bosonic direction under the fermion-parity conjugation, i.e. $\dim_{\text{super}} \mathfrak{g}_{\text{BPS}, (n, n)}^{\text{con}} = 2|0$. The 2-dimensional space is $\text{span}(h_n^{\text{Chev}}, h_n^{\text{tr}})$ (Remark 3.6); neither direction is projected out by DM integrality at the level of the CoHA — the Chevalley projection occurs only under the supertrace quotient to $\widehat{\mathfrak{sl}}(1|1)$, which is a downstream specialisation, not a DM-integrality input.*

Corollary: the super-PBW transport from $\text{CoHA}(Q_{\text{con}}, W_{\text{con}})$ to $U^+(\widehat{\mathfrak{gl}}(1|1))$ matches multiplicities with $\dim_{\text{super}}(\mathfrak{g}_{\text{BPS}}^{\text{con}})_\alpha = (0|1, 0|1, 2|0)$ on $((i+1, i), (i, i+1), (n, n))$ — odd real-root generators, two even imaginary-root generators (Chevalley + supertrace) per affine level. This is the super-dimension signature of $\widehat{\mathfrak{gl}}(1|1)^+$ ’s positive half, distinct from $\widehat{\mathfrak{sl}}_2^+$ (all real roots even, imaginary roots $(1|0)$) which appears only after the supertrace quotient.

Remark 3.3 (Two presentations of the BPS Lie algebra). **[Convention-dependent]** The BPS Lie algebra admits two equally valid presentations, operating at different scopes:

- (a) *Super reading* (Li–Yamazaki [arXiv:2003.08909](#) §8.3.6, Gaiotto–Rapčák [arXiv:1703.00982](#) corner $Y_{0,1,1}[\psi]$): $\mathfrak{g}_{\text{BPS}}^{\text{con}} = \widehat{\mathfrak{gl}}(1|1)^+$ as a super-Lie algebra, with two odd isotropic simple roots α_0, α_1 and Cartan generators $H_n = h_n^{(0)} - h_n^{(1)}$ (traceless, Chevalley) and $K_n = h_n^{(0)} + h_n^{(1)}$ (central; this is what distinguishes $\mathfrak{gl}(1|1)$ from $\mathfrak{sl}(1|1)$).
- (b) *Ungraded shadow* (MMNS motivic root-system reading, also [chapters/theory/cy_to_chiral.tex:7297](#)): $\mathfrak{g}_{\text{BPS}}^{\text{con}} \simeq \widehat{\mathfrak{sl}}_2^+$ after supertrace projection, which annihilates the central direction K and retains only the Chevalley Cartan H .

Both are theorems at their respective scopes. The super reading (a) is primary for the categorified/super CoHA; the ungraded shadow (b) is the semisimple quotient visible to motivic/numerical invariants. The 2-dim imaginary-root space at (n, n) is $\text{span}(H_n, K_n)$ in either reading; projection to H_n vs. retention of K_n is the only difference. Five routes (shuffle, Čech, VdB, RTT, chiral) probe different projections: routes (R2, R3, R8-motivic) pick out the Chevalley H ; routes (R1, R4, R5) pick out the central K via residue extraction at the coincidence locus.

Remark 3.4 (Cartan matrix convention). **[Convention-dependent]** The “super-Cartan matrix $C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ” frequently cited for $\widehat{\mathfrak{gl}}(1|1)$ is the *incidence matrix* of simple-root adjacency on the distinguished Dynkin diagram $\otimes - \otimes$, not the Kac–Moody generalised Cartan matrix (which is undefined for isotropic roots $a_{ii} = 0$). The symmetrised bilinear form on the affine root lattice has diagonal $(\alpha_i, \alpha_i) = 0$ (isotropy from Kac 1977 Thm. 2.4) and off-diagonal $(\alpha_0, \alpha_1) = \pm 1$ depending on the weight-basis parity convention (Frappat–Sciarrino–Sorba 2000 §2.41). All statements involving this matrix must name the convention.

Proposition 3.5 (Super-Lie bracket of $\mathfrak{g}_{\text{BPS}}^{\text{con}}$). **[Proposition]** *The affine Lie superalgebra $\widehat{\mathfrak{gl}}(1|1) = \mathfrak{gl}(1|1)[t, t^{-1}] \oplus \mathbb{C}K_0$ carries a single central element K_0 , not a family $\{K_n\}_{n \in \mathbb{Z}}$: the 2-cocycle of the affinisation is $\omega(xt^m, yt^n) = m\delta_{m+n,0} \text{str}(xy) K_0$ (Frappat–Sciarrino–Sorba 2000 §2.41; Kac 1977 Thm. 8.6). Under this cocycle and the Zeitlin [arXiv:0812.1961](#) normalisation of the invariant supersymmetric form, the super-Lie brackets on $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ read:*

$$\begin{aligned} \{e_m^{(a)}, e_n^{(a)}\}_+ &= 0 & (\text{isotropy; Remark 3.7}), \\ \{e_m^{(0)}, e_n^{(1)}\}_+ &= h_{m+n}^{\text{cross}} + m\delta_{m+n,0} \hbar K_0, \\ \{e_m^{(a)}, f_n^{(a)}\}_+ &= h_{m+n}^{(a)} + m\delta_{m+n,0} \hbar K_0, \\ [h_m^{(a)}, e_n^{(b)}] &= C_{ab} e_{m+n}^{(b)}, \\ [h_m^{(a)}, h_n^{(b)}] &= m C_{ab} \delta_{m+n,0} \hbar K_0, \\ [K_0, \cdot] &= 0 \quad (\text{central}), \end{aligned}$$

with $C_{ab} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ the incidence matrix (Remark 3.4) and cross-vertex Cartan $h_k^{\text{cross}} := h_k^{(0)} - h_k^{(1)}$ (Chevalley direction, Remark 3.3(a)). Centrality is forced: K_0 is the unique central element by Kac’s classification of affine Lie-superalgebraic central extensions (Kac 1977 Thm. 8.6); the would-be “ K_n for $n \neq 0$ ” is not a Lie-algebra element of $\widehat{\mathfrak{gl}}(1|1)$ but the Fourier mode $h_n^{\text{tr}} = h_n^{(0)} + h_n^{(1)}$ of the supertrace current (Remark 3.6). The mode- m cocycle contribution $m\delta_{m+n,0} \hbar K_0$ vanishes except on the zero Fourier-mode pair: the central extension is rank one, parametrised by K_0 alone, not by $m + n$.

Remark 3.6 (h^{tr} : central in $\widehat{\mathfrak{sl}}(1|1)$, not in $\widehat{\mathfrak{gl}}(1|1)$). **[Convention-dependent]** The supertrace current $h^{\text{tr}}(z) = h^{(0)}(z) + h^{(1)}(z)$ commutes with the Chevalley currents $e^{(a)}, f^{(a)}$ and generates an abelian Heisenberg ideal $[h_m^{\text{tr}}, h_n^{\text{tr}}] = m\delta_{m+n,0} \hbar K_0$ inside $\widehat{\mathfrak{gl}}(1|1)$; it is *not* central in $\widehat{\mathfrak{gl}}(1|1)$. In $\widehat{\mathfrak{sl}}(1|1) = \widehat{\mathfrak{gl}}(1|1)/\mathbb{C}h^{\text{tr}}[t, t^{-1}]$ this Heisenberg ideal is quotiented out and h^{tr} becomes central (Hatayama–Jimbo–Kuniba–Nakanishi [hep-th/9706096](#) Lemma 3.2). The super reading (Remark 3.3(a)) retains this abelian current as the K_n -labelled direction, coinciding with h_n^{tr} ; the ungraded shadow (b) annihilates it by supertrace projection. The CoHA-side 2-dim imaginary-root space at (n, n) is thus $\text{span}(h_n^{\text{Chev}}, h_n^{\text{tr}})$ in either reading; the routes differ only by whether they retain or project out h_n^{tr} .

Remark 3.7 (Isotropy at the pairing level, not merely rank). **[Proposition]** Pairing-level isotropy is a supertrace identity, not a rank count. In the $\mathfrak{gl}(1|1)$ weight basis (ϵ_1, ϵ_2) with $|\epsilon_1| = 0, |\epsilon_2| = 1$ and supersymmetric invariant form $(\epsilon_i, \epsilon_j) = (-1)^{|\epsilon_i|} \delta_{ij}$, the odd simple root $\alpha = \epsilon_1 - \epsilon_2$ pairs against itself by

$$(\alpha, \alpha) = (\epsilon_1, \epsilon_1) - 2(\epsilon_1, \epsilon_2) + (\epsilon_2, \epsilon_2) = 1 - 0 + (-1) = 0,$$

the supertrace cancellation of the even and odd diagonals. On the affine extension $\widehat{\mathfrak{gl}}(1|1) = \mathfrak{gl}(1|1)[t, t^{-1}] \oplus \mathbb{C}K_0$ the degree-shift raises $\alpha \mapsto \alpha_0 = \delta - \alpha$ and preserves isotropy: $(\alpha_0, \alpha_0) = (\delta, \delta) - 2(\delta, \alpha) + (\alpha, \alpha) = 0$ by $(\delta, \delta) = 0$ (affine imaginary direction) and $(\delta, \alpha) = 0$. This is the Kac 1977 Thm. 2.4 criterion evaluated on the supertrace form, not a statement about the rank of the Cartan matrix. The quiver absence of self-loops (which gives the shuffle kernel $\varphi_{\text{diag}}(u) = 1$) is a distinct datum: it encodes vanishing of the diagonal bond factor at the *module* level, whereas isotropy

encodes vanishing of the bilinear form at the *root* level. The two agree on the Klebanov–Witten quiver by the Davison–Meinhardt super-Poincaré polynomial identity ([arXiv:1601.02479](#) Thm. A), which bridges quiver-modular geometry to the supertrace pairing on $\mathfrak{g}_{\text{BPS}}$.

4. ROUTE R1 — DIRECT KLEBANOV–WITTEN SUPER-SHUFFLE

The super-shuffle algebra of the Klebanov–Witten quiver is

$$\text{Sh}_{\text{KW}}^{\text{super}} = \bigoplus_{(m,n) \in \mathbb{Z}_{\geq 0}^2} \mathbb{C}(z_1^{(0)}, \dots, z_m^{(0)} \mid z_1^{(1)}, \dots, z_n^{(1)})^{S_m \times S_n}$$

with product

$$(f \star g)(z, w) = \text{Sym}^{\text{super}} \left[f \cdot g \cdot \prod_{\text{cross arrows}} \varphi_{ab}(z_i^{(a)} - w_j^{(b)}) \right]$$

and bond factors given by Neguț’s [arXiv:1512.06473](#) rank-2 conifold kernel

$$(3) \quad \varphi^{0 \Rightarrow 0}(u) = \varphi^{1 \Rightarrow 1}(u) = 1, \quad \varphi^{0 \Rightarrow 1}(u) = \frac{(u + h_1)(u + h_2)}{u(u + h_1 + h_2)}, \quad \varphi^{1 \Rightarrow 0}(u) = \varphi^{0 \Rightarrow 1}(-u).$$

The diagonal bond vanishes because Q_{con} has no self-loops at either vertex; the cross bond is the toric-CY₃ specialisation of Neguț’s kernel (loc. cit., eq. (1.6)), identical to the Li–Yamazaki [arXiv:2003.08909](#) eq. (8.125) cross-arrow scalar for the conifold charge assignment, and equal to the rank-2 reduction of the Jordan kernel $(u + h_1)(u + h_2)(u + h_3)/u^3$ at $h_3 \rightarrow 0$ on the compact base (§1 and §R2). The numerator records the two KW arrows a_1, a_2 with T_{CY}^2 -weights (h_1, h_2) ; the denominator records the two Jacobi relations $\partial_{b_1} W = \partial_{b_2} W = 0$ with weights 0 and $h_1 + h_2$. The bond-factor product

$$(4) \quad \varphi^{0 \Rightarrow 1}(u) \varphi^{1 \Rightarrow 0}(u) = \varphi^{0 \Rightarrow 1}(u) \varphi^{0 \Rightarrow 1}(-u) = \frac{(u^2 - h_1^2)(u^2 - h_2^2)}{u^2(u^2 - (h_1 + h_2)^2)}$$

is the bond-factor norm; its zeros and poles record the diagonal residues computed in Prop. 4.3 below. The Kulish–Sklyanin R -matrix (9) factors as $R(u) = \varphi^{0 \Rightarrow 1}(u) \tilde{R}(u)$ with $\tilde{R}(u)$ the permutation-plus-identity core whose scalar unitarity holds (Arnaudon–Molev–Ragoucy [arXiv:math/0511481](#) §3.2).

Remark 4.1 (Rank-3 to rank-2 kernel specialisation). **[Proposition]** The rank-3 Jordan kernel $\varphi^{\mathbb{C}^3}(u) = (u + h_1)(u + h_2)(u + h_3)/u^3$ does *not* specialise at $h_3 \rightarrow 0$ to the conifold kernel (3) by a naive drop of the $(u + h_3)$ factor: the two denominators differ (u^3 for $\mathbb{C}^3$ versus $u(u + h_1 + h_2)$ for the conifold). The correct specialisation is the Neguț–Tsymbaliuk [arXiv:1404.5240](#) Prop. 4.1 rank-reduction procedure, which simultaneously drops a numerator factor and rearranges the denominator through a shifted moment map. Geometric source: $\mathbb{C}^3$ ’s three Jacobi relations all carry weight 0 under the cyclic potential $\text{tr}(x[y, z])$, giving $u \cdot u \cdot u = u^3$; the conifold’s two Jacobi relations carry weights 0 and $h_1 + h_2$, giving $u \cdot (u + h_1 + h_2)$.

Remark 4.2 (Supertrace reduction to $\widehat{\mathfrak{gl}}_1$ shuffle with fermionic direction). **[Proposition]** The KW super-shuffle reduces to the Schiffmann–Vasserot [arXiv:1202.2756](#) $\mathbb{C}^3$ shuffle tensored with a single Grassmann direction. Set $z_i^{\text{tot}} = z_i^{(0)}, \theta_j = z_j^{(1)} \in \Lambda^1$ and define the supertrace projection

$$\text{str}: \text{Sh}_{\text{KW}}^{\text{super}} \longrightarrow \text{Sh}_{\text{SV}}^{\mathbb{C}^3}[\theta],$$

sending the bond-factor pair $(\varphi^{0 \Rightarrow 1}(u), \varphi^{1 \Rightarrow 0}(u))$ to the rank-3 Jordan kernel at $h_3 = -h_1 - h_2$:

$$\text{str}(\varphi^{0 \Rightarrow 1}(u)) = \left. \frac{(u + h_1)(u + h_2)(u + h_3)}{u^3} \right|_{h_3 = -h_1 - h_2} = \varphi^{\mathbb{C}^3}(u)|_{\text{CY}}.$$

The map is a homomorphism: $\text{str}(f \star g) = \text{str}(f) \star_{\mathbb{C}^3} \text{str}(g)$. At Lie-algebra level this is the Remark 3.3 supertrace $\widehat{\mathfrak{gl}}(1|1) \rightarrow \widehat{\mathfrak{gl}}_1 \otimes \Lambda[\theta]$: the Chevalley direction $H = h^{(0)} - h^{(1)}$ becomes the $\mathbb{C}^3$ -shadow Heisenberg generator; the supertrace direction $K = h^{(0)} + h^{(1)}$ becomes the θ -direction; the odd isotropic simple roots reconstruct as $\theta \otimes e^{(\mathbb{C}^3)}$ in $\text{Sh}_{\text{SV}}^{\mathbb{C}^3}[\theta]$. Every KW super-shuffle computation admits a $\mathbb{C}^3$ -shadow cross-check through this projection.

Proposition 4.3 (Structure constants at low weight). **[Proposition]** *In the KW super-shuffle of eq. (3):*

- (1) (Cross product) $1_{(1,0)} \star 1_{(0,1)} = \varphi^{0 \Rightarrow 1}(z^{(0)} - z^{(1)})$.
 - (2) (Bosonic isotropy) $1_{(1,0)} \star 1_{(1,0)} = z_1^{(0)} + z_2^{(0)}$ (raw shuffle, since $\varphi^{0 \Rightarrow 0} = 1$); nonzero in $\text{Sh}_{\text{KW}}^{\text{super}}$, zero in the Davison critical-CoHA quotient $\text{Sh}_{\text{KW}}^{\text{super}}/I_{\text{wheel}}$ (Theorem 4.6), reproducing $(e_0^{(0)})^2 = 0$.
 - (3) (Fermionic isotropy) $1_{(0,1)} \star 1_{(0,1)} = 0$ identically by Koszul anti-symmetrisation, the shuffle-plane witness of Kac–Jantzen isotropy $(\alpha_1, \alpha_1) = 0$.
 - (4) (Imaginary-root structure constant)
- (5) $\mathbf{c}_{(1,1)}^{\text{R1}} = \frac{1}{2}(h_1 + h_2)^2 \cdot [\varphi^{0 \Rightarrow 1}(u) + \varphi^{0 \Rightarrow 1}(-u) - 2]_{u=0} / h_1^2 h_2^2 = \frac{1}{h_1 h_2} = (\varepsilon_1 \varepsilon_2)^{-1}$.

Proof of item 4. By the shuffle product, the super-anticommutator $\{1_{(1,0)}, 1_{(0,1)}\}_+$ in $\text{Sh}_{1,1}$ equals $\varphi^{0 \Rightarrow 1}(u) + \varphi^{0 \Rightarrow 1}(-u)$ at $u = z^{(0)} - z^{(1)}$, using $\varphi^{1 \Rightarrow 0}(v) = \varphi^{0 \Rightarrow 1}(-v)$. Substituting eq. (3) and expanding,

$$\varphi^{0 \Rightarrow 1}(u) = 1 + \frac{h_1 h_2}{u(u + h_1 + h_2)}, \quad \varphi^{0 \Rightarrow 1}(u) + \varphi^{0 \Rightarrow 1}(-u) - 2 = \frac{-2 h_1 h_2}{u^2 - (h_1 + h_2)^2},$$

finite at $u = 0$ (the double pole cancels). Evaluating,

$$[\varphi^{0 \Rightarrow 1}(u) + \varphi^{0 \Rightarrow 1}(-u) - 2]_{u=0} = \frac{2 h_1 h_2}{(h_1 + h_2)^2}.$$

The Nazarov $\frac{1}{2}$ -normalisation on the cross-vertex super-anticommutator (Prop. 3.5) combined with the Schiffmann–Vasserot normalisation $\hbar = h_1 h_2 / (h_1 + h_2)$ (line preceding Conjecture 9.10) gives $\mathbf{c}_{(1,1)}^{\text{R1}} = (h_1 h_2)^{-1} = (\varepsilon_1 \varepsilon_2)^{-1}$, as claimed. $\square$

Remark 4.4 $((\varepsilon_1 \varepsilon_2)^{-1}$ three ways). **[Proposition]** The $(1, 1)$ -imaginary-root structure constant of $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ equals $(\varepsilon_1 \varepsilon_2)^{-1}$ in three independent derivations:

- (a) *R1 super-shuffle* (this section, eq. (5)): residue of the Neguț kernel at the diagonal coincidence locus.
- (b) *R3 Van den Bergh tilting* (§6; displayed at the charge-lattice identification): Hall coefficient via integration over $\text{Ext}^1(S_0, S_1) = \mathbb{C}^2$ with Kontsevich ϕ_W -localisation on the critical locus.
- (c) *R4 RTT / Maulik–Okounkov* (§7; Remark 9.11): MO stable-envelope residue $\text{Res}_{u=0} R^{\text{MO}}(u) = (h_1 h_2 / (h_1 + h_2)) \cdot P^{\text{super}}$, contracted against $[\mathbb{P}^1] = [S_0] - [S_1]$.

All three descend from eq. (3) through distinct lenses, satisfying the three-path verification discipline.

Proposition 4.5 (Shuffle coproduct on Drinfeld-new currents). **[Proposition]** *The Hopf coproduct on $\text{Sh}_{\text{KW}}^{\text{super}}$ descended from the RTT super-Yangian coproduct on $Y(\widehat{\mathfrak{gl}}(1|1))$ under Gauss decomposition takes the Drinfeld-new form*

$$(6) \quad \Delta(e^{(a)}(u)) = e^{(a)}(u) \otimes 1 + \psi^{(a)}(u) \otimes e^{(a)}(u) + \sum_{k \geq 1} \hbar^k \Delta_k e^{(a)}(u), \quad a \in \{0, 1\},$$

with $e^{(a)}(u) = \sum_n e_n^{(a)} u^{-n-1}$ the Drinfeld-new simple-root currents, $\psi^{(a)}(u)$ the Cartan series, and Δ_k the $\hbar^k$ -corrections fixed by coassociativity (Ding–Frenkel *arXiv:q-alg/9608002* Thm. 4.3; super

version Zhang [arXiv:q-alg/9701011](#) Thm. 2.1). The first correction on the $(1, 1)$ -imaginary-root subspace reproduces eq. (5): $(\Delta_1 e^{(1)}(u))|_{(1,1)} = (h_1 h_2)^{-1} e^{(1)}(u) \otimes e^{(0)}(u)$.

Theorem 4.6 (Davison super-shuffle realisation of the KW CoHA). **[Proved elsewhere]** (Davison [arXiv:1311.6989](#) Thm. A at the KW pair $(Q_{\text{con}}, W_{\text{con}})$; super extension by Davison–Hennecart–Schlegel–Mejia [arXiv:2303.12592](#) Thm. B.) The critical CoHA of the resolved conifold embeds as the wheel quotient of the KW super-shuffle:

$$(7) \quad \text{CoHA}(Q_{\text{con}}, W_{\text{con}}) \hookrightarrow \text{Sh}_{\text{KW}}^{\text{super}} / I_{\text{wheel}},$$

with wheel ideal I_{wheel} generated by the bidegree- $(2, 1)$ and $(1, 2)$ Feigin–Odesskii wheel elements

$$(8) \quad \begin{aligned} W_{(2,1)}(z_1^{(0)}, z_2^{(0)}; z^{(1)}) &= (z_1^{(0)} - z^{(1)} + h_1)(z_2^{(0)} - z^{(1)} + h_2) - (1 \leftrightarrow 2), \\ W_{(1,2)}(z^{(0)}; z_1^{(1)}, z_2^{(1)}) &= (z^{(0)} - z_1^{(1)} + h_1)(z^{(0)} - z_2^{(1)} + h_2) - (1 \leftrightarrow 2), \end{aligned}$$

evaluated at the bond-factor zero pairs of eq. (3). Vanishing of $W_{(m,n)}$ inside the CoHA enforces the $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ Serre relations of Prop. 3.5. The super-PBW of Theorem 3.1 identifies $\text{CoHA}(Q_{\text{con}}, W_{\text{con}}) \cong \text{Sym}_{\text{super}}^{\boxplus+}(H(BC^\times)_{\text{vir}} \otimes \mathfrak{g}_{\text{BPS}}^{\text{con}})$ with the shuffle quotient under the Feigin–Odesskii dictionary.

Remark 4.7 (Quadratic super-Serre from Kulish–Sklyanin, not quartic). **[Proposition]** The self-Serre relation $(e_0^{(1)})^2 = 0$ (equivalently $\{e^{(1)}(u), e^{(1)}(v)\}_+ = 0$) descends from the Kulish–Sklyanin 1982 RTT relation directly, not as a Yamane postulate. Extract the $(12; 12)$ -block of $R_{12}(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R_{12}(u-v)$ with R the super- R -matrix of equation (9) and graded tensor $T_1(u)T_2(v) = T(u) \otimes T(v) \cdot (-1)^{|t_{ij}||t_{kl}|}$ (Arnaudon–Molev–Ragoucy [arXiv:math/0511481](#) §3 super version). The $(12; 12)$ -component isolates the $t_{12}(u)t_{12}(v)$ monomials; using $(2, 2; 2, 2) = 1 - \hbar/(u-v)$ (the minus sign from $P^{\text{super}}(e_2 \otimes e_2) = -e_2 \otimes e_2$ forces the fermionic contribution) and the cross-block entries, the RTT identity collapses to

$$(u - v - \hbar)(t_{12}(u)t_{12}(v) + t_{12}(v)t_{12}(u)) = 0,$$

that is $\{t_{12}(u), t_{12}(v)\}_+ = 0$ as formal series in u^{-1}, v^{-1} away from the pole $u - v = \hbar$ (Kulish–Sklyanin 1982 DOI:10.1007/BF01036128 Thm. 3.1; Arnaudon–Molev–Ragoucy [arXiv:math/0511481](#) Prop. 3.4 super). Gauss extraction $e^{(1)}(u) = d_1(u)^{-1}t_{12}(u)$ (Ding–Frenkel [arXiv:q-alg/9608002](#) super version, Zhang [arXiv:q-alg/9701011](#)) preserves the anticommutator since $d_1(u)$ commutes with $t_{12}(v)$ up to the Cartan-OPE scalar $\psi_1(u, v)$, absorbed into normalisation:

$$\{e^{(1)}(u), e^{(1)}(v)\}_+ = 0.$$

Specialisation $u \rightarrow v$ gives the integrated form $(e_0^{(1)})^2 = \frac{1}{2}\{e_0^{(1)}, e_0^{(1)}\}_+ = 0$, the quadratic super-Serre. The isotropy $a_{11} = 0$ is the *source* of this identity, not an independent postulate: if $a_{11} \neq 0$ the R -matrix $(2, 2; 2, 2)$ -entry would not be $1 - \hbar/u$ and the anticommutator would acquire a non-zero right-hand side. No Yamane cubic applies because the odd node has no adjacent even node of opposite parity. The wheel condition for the conifold shuffle algebra lives in bidegree $(2, 1)$ or $(1, 2)$ and is a $\mathbb{Z}/2$ -graded *quartic* wheel in total variable count; it is the shuffle-plane witness of the same RTT identity under the Ding–Frenkel map. The “cubic Serre via Neguț wheel” is the $\mathbb{C}^3$ Jordan-quiver wheel condition from Neguț 2015 §2.4, which does not apply here: the single vertex with three loops forces a cubic, while Klebanov–Witten’s two-vertex bipartite quiver forces a quadratic self-Serre and a separate quartic wheel.

5. ROUTE R2 — TWO-CHART ČECH DESCENT FOR THE COHA SHEAF

The two-chart atlas $\mathbf{Y} = U_+ \cup U_-$ supports Čech descent for the quasi-coherent CoHA sheaf, in the sense of Kontsevich–Soibelman [arXiv:1006.2706](#) §6.3. This is *not* a factorisation-algebra gluing: the two-chart open cover $\mathfrak{U}_0 = \{U_+, U_-\}$ fails the Weiss local-to-global axiom of Costello–Gwilliam [arXiv:2210.13036](#) Definition 6.1.6 — the two-point configuration $\{p_+, p_-\}$ with $p_+ \in U_+ \setminus U_-$ and

$p_- \in U_- \setminus U_+$ is contained in no member of $\mathfrak{U}_0$ (see the gluing chapter Part II, Proposition *prop:two-chart-not-weiss*). Factorisation-algebra locality requires the Weiss refinement $\mathfrak{D}^\sqcup$ by disjoint-union closures of disc neighbourhoods (Remark *rem:weiss-refinement-conifold*). The present section treats the two-chart Čech nerve, which recovers the weaker QC / sheaf-theoretic descent; residues and OPEs requiring two-point configurations with points in distinct charts (the $\widehat{\mathfrak{gl}}(1|1)$ fermionic Serre below) are computed on the Weiss refinement.

On each chart Costello–Li [arXiv:1605.09656](#) constructs the holomorphic Chern–Simons factorisation algebra $\mathcal{F}_\pm^{\text{hCS}}$ whose classical observables are polyvectors: $H^\bullet(\mathcal{F}_\pm^{\text{hCS}}) = \text{PV}^\bullet(U_\pm)$. Kontsevich–Tamarkin E_3 -formality together with HKR identifies the E_3 -algebra $\mathcal{F}_\pm^{\text{hCS}}$ with the Hochschild centre $HH^*(\Gamma_\pm)$ of the Ginzburg dga $\Gamma_\pm$ of the Jordan-triple quiver with cubic potential. Write

$$\mathcal{F}_\pm := \Phi_3^{FA}(\Gamma_\pm), \quad H^\bullet(\mathcal{F}_\pm) = \text{PV}^\bullet(U_\pm)$$

for the common chart-local E_3 -object, understanding the identification as the hCS-to-Hochschild quasi-isomorphism. Chart-wise CoHA $\text{CoHA}(U_\pm) = Y^+(\widehat{\mathfrak{gl}}_1)$ (Schiffmann–Vasserot [arXiv:1202.2756](#)) is the critical cohomology of the moduli stack $\mathcal{M}(U_\pm, W_\pm)$ over each chart.

Definition 5.1 (Fourier–Mukai transition kernel). Let $\phi_{+-} : U_+ \cap U_- \xrightarrow{\sim} U_+ \cap U_-$ denote the biholomorphism $(z, u_+, v_+) \mapsto (z^{-1}, zu_+, zv_+)$ of equation (1), and let $\Gamma_{\phi_{+-}} \subset (U_+ \cap U_-) \times (U_+ \cap U_-)$ be its graph. The Čech transition kernel is the structure sheaf of the graph twisted by the normal-bundle class of the $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ fibre:

$$K_{+-} = \mathcal{O}_{\Gamma_{\phi_{+-}}} \otimes \pi_1^* \mathcal{O}_{\mathbb{P}^1}(-1)|_{\mathbb{C}^\times},$$

with π_1 the first projection $(U_+ \cap U_-) \times (U_+ \cap U_-) \rightarrow U_+ \cap U_-$. The induced derived pullback $L\phi_{+-}^*$ on polyvectors decomposes the z -weight additively into a base factor (z^{-1} from the $\mathbb{P}^1$ -transition $\tilde{z} = z^{-1}$) and fibre factors (z^{-a}, z^{-b} from the fibre-weight shifts on $\partial_{u_-}, \partial_{v_-}$ coming from $u_- = zu_+, v_- = zv_+$):

$$T_{+-}^* : f(z, u_+, v_+) \partial_z^\alpha \partial_{u_+}^a \partial_{v_+}^b \mapsto f(z^{-1}, zu_+, zv_+) z^{-(1+a+b)} \partial_{\tilde{z}}^\alpha \partial_{u_-}^a \partial_{v_-}^b.$$

Remark 5.2 (FM kernel vs. Bridgeland–Chen flop kernel). The kernel K_{+-} is the analytic chart-transition FM kernel on the non-exceptional locus, not the global Bridgeland–Chen flop kernel of [arXiv:math/0009053](#), [math/0209342](#). The two kernels agree on $U_+ \cap U_-$ where the flop is an isomorphism; the global flop kernel lives on a birational fibre product and has additional support over the exceptional $\mathbb{P}^1$. When quoting “FM kernel” one must disambiguate.

Theorem 5.3 (Čech QC descent for the CoHA sheaf). **[Proved elsewhere]** (Kontsevich–Soibelman [arXiv:1006.2706](#) §6.3; the $(\infty, 1)$ -categorical ambient is Lurie HA §5.5.4 of presentable stable symmetric-monoidal categories.) *In the $(\infty, 1)$ -category $\text{QCoh}^\otimes(\mathbf{Y})$ the derived pullback square*

$$\mathcal{F}_{\mathbf{Y}} \simeq \mathcal{F}_+ \times_{\mathcal{F}_+|_{U_+ \cap U_-}}^h \mathcal{F}_- = \text{hocolim}(\mathcal{F}_- \xleftarrow{T_{+-}} \mathcal{F}_+|_{U_+ \cap U_-} \xrightarrow{\text{id}} \mathcal{F}_+)$$

assembles a QC-CoHA sheaf on $\mathbf{Y}$ with $H^\bullet(\mathcal{F}_{\mathbf{Y}}) = \text{PV}^\bullet(\mathbf{Y})$. The two-chart nerve terminates at pair overlaps (no triple overlap exists), so the pullback square presentation is equivalent to the full Čech hocolim. The assembly does not produce an E_3 -factorisation algebra on $\mathbf{Y}$ in the Costello–Gwilliam sense (the cover is not Weiss); the corresponding factorisation algebra is constructed on the Weiss refinement $\mathfrak{D}^\sqcup$ by Francis–Gaitsgory-type descent ([arXiv:1103.5803](#) §3.6; Francis [arXiv:1303.0305](#) for the higher- E_n extension; Lurie HA §5.5.4) and agrees with $\mathcal{F}_{\mathbf{Y}}$ on the shared QC-sheaf content.

The glued QC-sheaf carries a $\mathbb{Z}/2$ -grading from the $\mathcal{O}(-1)$ -cocycle. The sign source separates cleanly: the $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ fibre is CY-balanced ($c_1(T\mathbb{P}^1) + c_1(\text{fibre}) = 0$), so the fibre transitions preserve the CY volume form $\Omega_{\mathbf{Y}}$ identically; the sign comes from the $\mathbb{P}^1$ base, where the orientation class $[\mathbb{P}^1] \in H^2(\mathbb{P}^1, \mathbb{Z})$ reverses under $z \leftrightarrow \tilde{z} = z^{-1}$ (see Kontsevich–Soibelman [arXiv:1006.2706](#) §6.3).

on chart-gluing signs for CoHA on toric CY_3). The resulting Koszul sign $\sigma = -1$ acts on cross-chart Hall products. Even/odd combinations

$$E^{(0)} = e_0^{(+)} + e_0^{(-)}, \quad E^{(1)} = e_0^{(+)} - e_0^{(-)}$$

are introduced as objects of the QC-CoHA sheaf. Their fermionic Serre relation $\{E^{(1)}, E^{(0)}\}_+ = K^{(0)}$ (central) requires a two-point residue as $z_1 \rightarrow z_2$ with z_1, z_2 chosen in distinct charts; this configuration is invisible to the two-chart Čech nerve and is computed on the Weiss refinement $\mathfrak{D}^\sqcup$ (Remark *rem:weiss-refinement-conifold* of gluing Part 5), where disjoint disc neighbourhoods around $z_1 \in U_+$ and $z_2 \in U_-$ are available. On $\mathfrak{D}^\sqcup$ the Beilinson–Drinfeld chiral-bracket residue $\mu: j_* j^*(\mathcal{F}_Y \boxtimes \mathcal{F}_Y) \rightarrow \Delta_! \mathcal{F}_Y$ (*Chiral Algebras* §3.4; Francis–Gaitsgory [arXiv:1103.5803](#) Thm 3.6.2) computes the Jordan-shuffle difference that assembles the fermionic Serre of $\widehat{\mathfrak{gl}}(1|1)$ from the two bosonic $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks.

The $\varepsilon_3 \rightarrow 0$ specialisation on the compact base collapses the chart- $\mathbb{C}^3$ shuffle kernel numerator from the rank-3 form $(u + \varepsilon_1)(u + \varepsilon_2)(u + \varepsilon_3)/u^3$ to the rank-2 form $(u + \varepsilon_1)(u + \varepsilon_2)/[u(u + \varepsilon_1 + \varepsilon_2)]$ of Neguţ, reproducing eq. (3).

6. ROUTE R3 — VAN DEN BERGH TILTING

The geometric input is the tilting generator $T = \mathcal{O}_Y \oplus \mathcal{O}_Y(1)$ of Theorem 2.1, with $\mathcal{O}_Y(1) = \pi^* \mathcal{O}_{\mathbb{P}^1}(1)$ pulled back along the projection $\pi: Y \rightarrow \mathbb{P}^1$ (Bondal–Orlov [arXiv:math/0206295](#) §3; Van den Bergh [arXiv:math/0211064](#) Prop. 3.2.10). The derived Morita adjoint pair (L, R) of Theorem 2.1 gives quasi-inverse equivalences $D^b(\text{Coh } Y) \simeq D^b(J(Q_{\text{con}}, W_{\text{con}})\text{-mod})$ in full generality: compact generation of T is Bondal–Van den Bergh [arXiv:math/0204218](#) Thm. 3.1.1 applied to the Beilinson pair $(\mathcal{O}_{\mathbb{P}^1}, \mathcal{O}_{\mathbb{P}^1}(1))$ on $\mathbb{P}^1$ lifted by π ; vanishing of positive Ext reduces to $H^{\geq 1}(\mathbb{P}^1, \mathcal{O}(-1)) = 0$ via Leray. Under $L = (-) \otimes_J^L T$ the two vertex-simples of $J(Q_{\text{con}}, W_{\text{con}})$ map to the exceptional twists

$$L(S_0) = \mathcal{O}_{\mathbb{P}^1}(-1)[1], \quad L(S_1) = \mathcal{O}_{\mathbb{P}^1},$$

as pure complexes supported on the exceptional $\mathbb{P}^1 \subset Y$ (Bridgeland [arXiv:math/0502050](#) §5; Szendrői [arXiv:0705.3419](#) §2.2); under $R = R\text{Hom}_Y(T, -)$ the closed-point skyscraper $\mathcal{O}_p$, $p \in Y$, decomposes to a bounded J -complex whose class in $K_0(J) = \mathbb{Z}\langle [S_0], [S_1] \rangle$ is $[S_0] + [S_1]$. The exceptional $\mathbb{P}^1$ carries class $[\mathbb{P}^1] = [S_0] - [S_1]$ via $0 \rightarrow \mathcal{O}_{\mathbb{P}^1}(-1) \rightarrow \mathcal{O}_{\mathbb{P}^1} \rightarrow \mathcal{O}_p \rightarrow 0$.

Davison–Hennecart–Schlegel-Mejia [arXiv:2303.12592](#) Theorem A transports the bialgebra structure along L : the derived Morita equivalence upgrades to a *bialgebra isomorphism* of critical cohomological Hall algebras

$$\text{CoHA}^{\text{crit}}(\text{Coh } Y, 0) \xrightarrow{\sim} \text{CoHA}^{\text{crit}}(J(Q_{\text{con}}, W_{\text{con}})\text{-mod}, \varphi_{W_{\text{con}}}),$$

with the Jacobi-side critical cocycle $\varphi_{W_{\text{con}}}$ matching the zero-potential smooth geometric side (smoothness of Y : critical locus = full moduli). Multiplication is categorical Hall convolution on both sides; comultiplication is the Davison nilpotent-tangent decomposition, upgrading the 2-Calabi–Yau isomorphism of Davison [arXiv:1903.02860](#) to the CY-3 Jacobi setting via the Davison–Meinhardt super-PBW decomposition [arXiv:1601.02479](#) / [arXiv:2303.12592](#) Thm. B, whose common image

$$\text{Sym}_{\text{super}}^{\boxplus+}(H(B\mathbb{C}^\times)_{\text{vir}} \otimes \mathfrak{g}_{\text{BPS}}^{\text{con}})$$

identifies the super BPS Lie algebra $\mathfrak{g}_{\text{BPS}}^{\text{con}} = \widehat{\mathfrak{gl}}(1|1)^+$ on both sides of the equivalence (Theorem 3.1, Remark 3.3).

In the $(d_0, d_1) \leftrightarrow (D_0, D_2)$ coordinates of Remark 3.3, D_0 is the point-class (D0-brane charge, even) and D_2 is the $\mathbb{P}^1$ -wrapping class (D2-brane charge, odd):

$$(d_0, d_1) \longleftrightarrow \left(\frac{1}{2}(D_0 + D_2), \frac{1}{2}(D_0 - D_2)\right), \quad [\mathbb{P}^1] = [S_0] - [S_1].$$

The Hall coefficient at charge $(1, 1)$, computed by equivariant localisation over $\text{Ext}^1(S_0, S_1) = \mathbb{C}^{\oplus 2}$ (the $(0, 1)$ -row of the Ext-table following (2)) against the critical cocycle $\varphi_{W_{\text{con}}}$, evaluates to $(\varepsilon_1 \varepsilon_2)^{-1}$ with $\varepsilon_3 = -\varepsilon_1 - \varepsilon_2$ (CY-3 balanced Serre condition), matching the $(1, 1)$ -coefficient of Route R1 (Section 4) and Route R2 (Section 5).

6.1. Minimal-model structure on the Ginzburg dg-algebra. Transport of the Hall product along L is controlled by the A_∞ -structure on $\text{Perf}(\Gamma_{\text{con}})$, where $\Gamma_{\text{con}} = \Gamma(Q_{\text{con}}, W_{\text{con}})$ is the Ginzburg dg-algebra of Theorem 2.1 (Ginzburg [arXiv:math/0612139](#) Thm. 3.2.8; Keller–Van den Bergh [arXiv:0906.0761](#) Thm. 6.3). Generators sit in degrees $[-2, 0]$: arrows a_1, a_2, b_1, b_2 in degree 0, cyclic duals $a_1^*, a_2^*, b_1^*, b_2^*$ in degree -1 , loop generators t_0, t_1 at the two vertices in degree -2 , with differential $\partial a_i^* = \partial_{a_i} W_{\text{con}}$ on the (-1) -layer and $\partial t_j = \sum_{a:j \rightarrow j} [a, a^*]$ on the (-2) -layer (Ginzburg [arXiv:math/0612139](#) eqn. 5.3.1). The four Jacobi relations (2) are the degree-0 cohomology of ∂ .

Kadeishvili’s minimal-model transfer (Kadeishvili 1982; Kontsevich [arXiv:q-alg/9709040](#) §6.4) realises $H^\bullet(\Gamma_{\text{con}}) = J(Q_{\text{con}}, W_{\text{con}})$ as an A_∞ -algebra $(J, m_2, m_3, m_4, \dots)$ with each m_k computed from sums over planar binary trees of ∂ -splittings on Γ_{con} . The L_∞ -shadow on the cyclic-invariant quotient $\text{HC}_\bullet(\Gamma_{\text{con}})$ carries the Gerstenhaber bracket of degree $1 - d = -2$ (Pantev–Toën–Vaquié–Vezzosi [arXiv:1111.3209](#) shift law at $d = 3$: E_1 cyclic on A , E_2 Gerstenhaber on the Hochschild centre). The Maurer–Cartan element

$$\mu = m_2 + \hbar m_3 + \hbar^2 m_4 + \dots \in C^\bullet(\Gamma_{\text{con}}, \Gamma_{\text{con}})[[\hbar]]$$

encodes the deformation of the classical Jacobi product m_2 into the critical Hall product; closure $[\mu, \mu] = 0$ is the L_∞ Jacobi identity on the Hochschild cochain complex and is equivalent, via the Davison–Meinhardt super-PBW [arXiv:1601.02479](#), to the co-associativity of the Hall coproduct.

Kontsevich–Tamarkin formality at $d = 3$ (Kontsevich [arXiv:q-alg/9709040](#) Main Thm.; Tamarkin [arXiv:math/9803025](#) for the Gerstenhaber upgrade) in the purely-even gauge gives an L_∞ -quasi-isomorphism $T_{\text{poly}}(X) \simeq C^\bullet(\mathcal{O}_X, \mathcal{O}_X)$ via Kontsevich’s $\mathcal{G} \simeq \mathcal{G}_\infty$ operadic resolution, determined up to a $\text{GRT}_1(\mathbb{Q})$ -torsor (Kontsevich [arXiv:q-alg/9709040](#) §4.6). In the super extension to $\widehat{\mathfrak{gl}}(1|1)$, Ginzburg–Schedler [arXiv:0807.0174](#) Thm. 1 establishes super-formality *only to order E_2* : Gerstenhaber-bracket closure persists, but the E_3 enhancement to cyclic operads on the super-Hochschild complex is obstructed by the super-wheel d_{super}^{abc} at 2-loop and higher (Costello–Gwilliam vol. 2 [arXiv:2210.13036](#) §12.4 one-loop analysis). Consequently the A_∞ -structure on $H^\bullet(\Gamma_{\text{con}})$ is E_2 -rigid but its E_3 lift carries a nontrivial cocycle obstruction; the bialgebra transport along L uses only the E_2 layer, which supplies exactly the input for the Davison–Hennecart–Schlegel-Mejia theorem.

6.2. Hall-product compatibility diagram.

Proposition 6.1 (Compatibility of Hall convolution with tilting transport). **[Proved elsewhere]** Let $\mathcal{M}_{\mathbf{Y}}^{d_0, d_1}$ and $\mathcal{M}_J^{d_0, d_1}$ denote the moduli stacks of coherent sheaves on $\mathbf{Y}$ and finite-dimensional $J(Q_{\text{con}}, W_{\text{con}})$ -modules of charge $(d_0, d_1) \in K_0 = \mathbb{Z}_{\geq 0}^2$ respectively. The derived Morita equivalence L lifts to an isomorphism of moduli stacks $\mathcal{M}_{\mathbf{Y}}^{d_0, d_1} \simeq \mathcal{M}_J^{d_0, d_1}$ (Bondal–Van den Bergh [arXiv:math/0204218](#) §6), and the Hall-product diagram

$$\begin{array}{ccc} H_c^\bullet(\mathcal{M}_{\mathbf{Y}}^{d_0, d_1}) \otimes H_c^\bullet(\mathcal{M}_{\mathbf{Y}}^{d'_0, d'_1}) & \xrightarrow{m_{\mathbf{Y}}^{\text{Hall}}} & H_c^\bullet(\mathcal{M}_{\mathbf{Y}}^{d_0+d'_0, d_1+d'_1}) \\ \downarrow L_* \otimes L_* & & \downarrow L_* \\ H_c^\bullet(\mathcal{M}_J^{d_0, d_1}) \otimes H_c^\bullet(\mathcal{M}_J^{d'_0, d'_1}) & \xrightarrow{m_J^{\text{Hall}}} & H_c^\bullet(\mathcal{M}_J^{d_0+d'_0, d_1+d'_1}) \end{array}$$

commutes, with $m_{\mathbf{Y}}^{\text{Hall}}$ the Kontsevich–Soibelman [arXiv:1006.2706](#) §6.1 categorical Hall convolution for coherent sheaves on $\mathbf{Y}$ and m_J^{Hall} the Szendrői [arXiv:0705.3419](#) / Schiffmann–Vasserot

*arXiv:1202.2756 §4.2 Hall convolution for Jacobi-algebra modules against the critical cocycle $\varphi_{W_{\text{con}}}$. The same square upgrades to compatibility of the Davison nilpotent-tangent comultiplications (Davison *arXiv:1903.02860* Thm. 1.1), yielding the bialgebra isomorphism *arXiv:2303.12592* Thm. A.*

The compatibility is the stack-level incarnation of Theorem 2.1: the 3-term correspondence

$$\mathcal{M}_{\mathbf{Y}}^{d_0, d_1} \times \mathcal{M}_{\mathbf{Y}}^{d'_0, d'_1} \xleftarrow{p} \text{Ext}_{\mathbf{Y}}^1 \xrightarrow{q} \mathcal{M}_{\mathbf{Y}}^{d_0+d'_0, d_1+d'_1}$$

defining $m_{\mathbf{Y}}^{\text{Hall}} = q_* p^*$ on extension classes (Kontsevich–Soibelman *arXiv:1006.2706* §6) pulls back along L to the Jacobi Ext^1 -correspondence on J -modules, with the critical cocycle pulling back to $\varphi_{W_{\text{con}}}$. Serre-duality compatibility on the Davison coproduct (the CY-3 shift [3] on both sides) follows from the bimodule self-duality $\Gamma_{\text{con}} \simeq R\text{Hom}_{\Gamma^e}(\Gamma_{\text{con}}, \Gamma^e)[3]$ (Ginzburg *arXiv:math/0612139* Thm. 3.2.8; quoted in Theorem 2.1) matched under L to $\omega_{\mathbf{Y}}[3] = \mathcal{O}_{\mathbf{Y}}[3]$ on the geometric side via the adjunction (L, R) with shift.

Remark 6.2 (Triangulation of the $(1, 1)$ -coefficient across three routes). **[Convention-dependent]** The $(1, 1)$ -Hall coefficient $(\varepsilon_1 \varepsilon_2)^{-1}$ arises on three distinct categorical strata, each computing the same invariant: (i) as the residue of the super-shuffle bond factor $\varphi^{0 \Rightarrow 1}(u) = (u+h_1)(u+h_2)/[u(u+h_1+h_2)]$ at the coincidence locus $u = 0$ (Route R1 super-shuffle, Section 4, eq. (5)), under the CY-balanced specialisation $\varepsilon_3 = -\varepsilon_1 - \varepsilon_2$; (ii) as the cocycle residue of the two-chart $\widehat{\mathfrak{gl}}_1$ -stalk gluing ϕ_{+-}^{Weiss} on the Weiss refinement $\mathfrak{D}^{\sqcup}$ of the conifold Čech cover (Route R2 Čech gluing, Section 5), computed via the chiral-bracket residue $\mu : j_* j^*(\mathcal{F}_{\mathbf{Y}} \boxtimes \mathcal{F}_{\mathbf{Y}}) \rightarrow \Delta_! \mathcal{F}_{\mathbf{Y}}$ (Beilinson–Drinfeld *Chiral Algebras* §3.4); (iii) as the localised equivariant integral over $\text{Ext}^1(S_0, S_1) = \mathbb{C}^{\oplus 2}$ against the critical cocycle $\varphi_{W_{\text{con}}}$ (Route R3, the present section), transported along L by Proposition 6.1. The three computations agree because each is a shadow of the Kontsevich–Soibelman cocycle $\phi^+(u)$ for the conifold UV positive half: shuffle bond factor, Weiss cocycle restriction, Jacobi critical cocycle. The identification across routes is the MO-bypass lemma for the conifold (chiral-side manifestation of the Maulik–Okounkov R -matrix as a gluing-cocycle residue, Schiffmann–Vasserot *arXiv:1202.2756* §3 read through *arXiv:1006.2706* §5.4).

Remark 6.3 (Bridgeland-stability compatibility across chambers). **[Proved elsewhere]** The tilting generator $T = \mathcal{O}_{\mathbf{Y}} \oplus \mathcal{O}_{\mathbf{Y}}(1)$ is a Bridgeland-stable object for the standard heart $\text{Coh } \mathbf{Y} \subset D^b(\text{Coh } \mathbf{Y})$ in Bridgeland’s stability-condition space $\text{Stab}(\mathbf{Y})$ (Bridgeland *arXiv:math/0502050* §5, explicitly for the resolved conifold and its flop). Across walls of marginal stability, the Jacobi-side heart $J(Q_{\text{con}}, W_{\text{con}})\text{-mod} \subset D^b(J\text{-mod})$ is a distinct tilt — the Szendrői non-commutative chamber of Theorem 3.1 — related to the geometric heart by a Seidel–Thomas spherical twist $T_{\mathcal{O}_{\mathbb{P}^1}(-1)} \in \text{Auteq } D^b(\text{Coh } \mathbf{Y})$ (*arXiv:math/0502050* Thm. 1.1). The DT partition function Z^{NC} of *arXiv:1107.5017* Theorem 3.1 is the non-commutative-chamber specialisation of the universal Joyce–Song generating series; wall-crossing to the large-volume chamber reproduces the Maulik–Nekrasov–Okounkov–Pandharipande Gromov–Witten series via the MNOP conjecture (Maulik–Nekrasov–Okounkov–Pandharipande *arXiv:math/0312059* §1; Toda *arXiv:0806.0062* for the DT/PT equivalence; proved on the conifold by Morrison–Mozgovoy–Nagao–Szendrői *arXiv:1107.5017* Thm. 2). The VdB tilting transports the CoHA bialgebra faithfully between the two chambers; the Davison–Hennecart–Schlegel–Mejia isomorphism is a chamber-independent statement, with the Koszul-sign cocycle $\sigma = \pm 1$ on cross-chamber Hall products controlled by the $c_1(\mathcal{O}_{\mathbf{Y}}(1))$ -twist absorbed into L (Section 4 above on the Koszul $\mathbb{Z}/2$ -grading of the glued QC-sheaf).

7. ROUTE R4 — RTT SUPER-YANGIAN

The route realises the CoHA of the conifold as the RTT super-Yangian $Y^{\text{RTT}}(\widehat{\mathfrak{gl}}(1|1))$ cut on equivariant slices indexed by the Ω -background parameters $(\varepsilon_1, \varepsilon_2, \varepsilon_3)$ of Section ???. The match to

Routes R1–R3, R5 is a theorem at the leading $(1, 1)$ -weight (Theorem 7.1 and Proposition ??) and conjectural at subleading weight (Conjecture 9.10, Remark 9.11).

The super- R -matrix in its graded basis. On $V = \mathbb{C}^{1|1}$ with basis (e_1, e_2) , $|e_1| = 0$, $|e_2| = 1$, the graded flip

$$P^{\text{super}}(e_i \otimes e_j) = (-1)^{|e_i||e_j|} e_j \otimes e_i$$

acts on the ordered basis $(e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2)$ with matrix $\text{diag}(1, 0, 0, -1)$ on the diagonal pairs and the off-diagonal flip $e_1 \otimes e_2 \leftrightarrow e_2 \otimes e_1$. Kulish–Sklyanin 1982 write

$$(9) \quad R(u) = \text{Id} + \frac{\hbar}{u} P^{\text{super}} = \begin{pmatrix} 1 + \frac{\hbar}{u} & 0 & 0 & 0 \\ 0 & 1 & \frac{\hbar}{u} & 0 \\ 0 & \frac{\hbar}{u} & 1 & 0 \\ 0 & 0 & 0 & 1 - \frac{\hbar}{u} \end{pmatrix},$$

with $(22; 22)$ -entry $1 - \hbar/u$ from $P^{\text{super}}(e_2 \otimes e_2) = -e_2 \otimes e_2$. Schur’s lemma on the two-dimensional fixed-weight space of the $\mathfrak{gl}(1|1)$ -action on $V^{\otimes 2}$ pins (9) up to overall scalar at rational order.

Theorem 7.1 (Super–Yang–Baxter by direct matrix elements). **[Theorem]** *The super- R -matrix (9) satisfies*

$$R_{12}(u) R_{13}(u+v) R_{23}(v) = R_{23}(v) R_{13}(u+v) R_{12}(u) \quad \text{on } V^{\otimes 3},$$

with the graded tensor product $(A \otimes B)(x \otimes y) = (-1)^{|B||x|} Ax \otimes By$.

Proof. Write $R(u) = \text{Id} + (\hbar/u) P^{\text{super}}$ and abbreviate P_{ij} for the graded flip on factors i, j of $V^{\otimes 3}$. Expanding YBE in powers of $\hbar$ yields three identities:

$$\begin{aligned} \hbar^1: \quad & \frac{1}{u} P_{12} + \frac{1}{u+v} P_{13} + \frac{1}{v} P_{23} = \frac{1}{v} P_{23} + \frac{1}{u+v} P_{13} + \frac{1}{u} P_{12}, \\ \hbar^2: \quad & \frac{1}{u(u+v)} P_{12} P_{13} + \frac{1}{uv} P_{12} P_{23} + \frac{1}{(u+v)v} P_{13} P_{23} \\ & = \frac{1}{v(u+v)} P_{23} P_{13} + \frac{1}{uv} P_{23} P_{12} + \frac{1}{u(u+v)} P_{13} P_{12}, \\ \hbar^3: \quad & \frac{1}{u(u+v)v} P_{12} P_{13} P_{23} = \frac{1}{v(u+v)u} P_{23} P_{13} P_{12}. \end{aligned}$$

The $\hbar^1$ identity is trivial. The $\hbar^3$ identity, clearing the common denominator $u(u+v)v$, is the graded braid relation $P_{12} P_{13} P_{23} = P_{23} P_{13} P_{12}$, the S_3 presentation $\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2$ for the graded symmetric group on three letters. The $\hbar^2$ identity, after multiplying through by $uv(u+v)$, reduces on a basis vector $e_i \otimes e_j \otimes e_k$ to the matching of two $\mathbb{Z}$ -linear combinations of the six permutations of (e_i, e_j, e_k) weighted by $v, u, u+v$; overall sign $(-1)^{|e_i||e_j|+|e_j||e_k|+|e_i||e_k|}$ is symmetric (counts graded inversions modulo 2), and the scalar polynomial matching reduces to the unsuper $\mathfrak{gl}_2$ YBE (Yang 1967) on the diagonal weight- $(1, 1)$ subrepresentation. $\square$

Remark 7.2 (What the direct proof replaces). Earlier drafts invoked Mukhin–Varchenko [arXiv:math/0303045](#) or the rank-2 case of Arnaudon–Crampé–Doikou–Frappat–Ragoucy [arXiv:math/0511481](#); both are useful cross-references but are not indispensable for the verification. Kulish–Sklyanin 1982 remains the primary source for the ansatz (9); Theorem 7.1 verifies YBE from the graded flip alone. The super–Yangian presentation (Yamane [arXiv:math/9603017](#) Thm. 6.4.1) is derived downstream from the RTT relation built on this R -matrix.

RTT relation with graded tensor (Arnaudon–Molev–Ragoucy [arXiv:math/0511481](#) Thm. 3.2 super):

$$R_{12}(u-v) T_1(u) T_2(v) = T_2(v) T_1(u) R_{12}(u-v).$$

Gauss decomposition $T(u) = F(u) D(u) E(u)$ with $D(u) = \text{diag}(d_1(u), d_2(u))$ yields Drinfeld–new currents

$$e^{(1)}(u) = d_1(u)^{-1} t_{12}(u), \quad f^{(1)}(u) = t_{21}(u) d_1(u)^{-1}, \quad \psi(u) = d_2(u) d_1(u)^{-1},$$

with Ding–Frenkel [arXiv:q-alg/9608002](#) super-version (cognate: Zhang [arXiv:q-alg/9701011](#) “Super Yangian Double $DY(\mathfrak{gl}(1|1))$ and its Gauss Decomposition”; the earlier-cited [arXiv:q-alg/9603024](#) is a different paper and was an erroneous citation).

8. ROUTE R5 — CHIRAL VERTEX-ALGEBRA REALISATION

Remark 8.1 (Corner assignment is a Li–Yamazaki synthesis, not a Gaiotto–Rapčák theorem). **[Conjectural]** Gaiotto–Rapčák [arXiv:1703.00982](#) construct $Y_{N_1, N_2, N_3}[\psi]$ as the factorisation algebra on the trivalent junction of three holomorphic-topological BV theories: Costello 5D holomorphic Chern–Simons on $\mathbb{R} \times \mathbb{C}_i^2$ ($i = 1, 2, 3$) with BV action

$$S_i = \frac{1}{2\pi i} \int_{\mathbb{R} \times \mathbb{C}_i^2} \Omega_i \wedge \text{tr} \left(A_i \wedge \bar{\partial} A_i + \frac{2}{3} A_i^{\wedge 3} \right),$$

Ω_i a holomorphic volume form on $\mathbb{C}_i^2$, $A_i \in \Omega^{0, \bullet}(\mathbb{C}_i^2; \mathfrak{gl}(N_i))$, glued along half-space interfaces carrying $\mathfrak{gl}(N_i|N_j)$ -valued defect fields (Costello [arXiv:1303.2632](#) §2; Costello–Gwilliam [arXiv:2210.13036](#) §4 for the factorisation-algebra formalism). GR Theorem 3 (§4) identifies the degenerate vertex $Y_{0,0,N}[\psi] = \mathcal{W}_N[\psi]$ as a truncation of $\mathcal{W}_{1+\infty}[\lambda]$ at specific ϵ_i -ratios. The assignment of the resolved conifold geometry to $(N_1, N_2, N_3) = (0, 1, 1)$ is *not* stated in [arXiv:1703.00982](#); it is the *Li–Yamazaki synthesis* [arXiv:2003.08909](#) §8.3.6, mapping the Klebanov–Witten quiver to the $(0, 1, 1)$ corner at $\psi = -\epsilon_1/\epsilon_3$ via the toric-CY₃-to-corner-label dictionary. The identification

$$V(\widehat{\mathfrak{gl}}(1|1))_\psi^{qv} \stackrel{?}{=} Y_{0,1,1}[\psi]$$

is **[Conjectural]**; primary-literature controls are: proved only on the free-field degeneration $\epsilon \rightarrow 0$ (Bershtein–Feigin–Merzon [arXiv:1512.08779](#)) and on the self-dual locus $\epsilon_1 + \epsilon_2 = 0$. The further identification with the Zamolodchikov $\mathcal{W}$ -algebra $\mathcal{W}(\mathfrak{gl}(1|1))_\psi$ runs through Creutzig–Gaiotto [arXiv:1706.02834](#) logarithmic extension and is open at non-generic ψ .

Proposition 8.2 (Holomorphic-topological BV decomposition of 5D hCS observables). **[Proved elsewhere]** (Costello–Gwilliam [arXiv:2210.13036](#) §4.4–§4.8 for the holomorphic-topological factorisation; Costello [arXiv:1303.2632](#) §2 for the 5D hCS theory on $\mathbb{R} \times \mathbb{C}^2$.) *The classical BV complex of 5D hCS on $\mathbb{R} \times \mathbb{C}^2$ with gauge algebra $\mathfrak{gl}(N)$ carries a product decomposition of observables*

$$\text{Obs}_{\mathbb{R} \times \mathbb{C}^2}^{\text{cl}} \simeq \text{Obs}_{\mathbb{R}}^{1\text{D top}} \otimes \text{Obs}_{\mathbb{C}^2}^{4\text{D hol}},$$

with $\text{Obs}_{\mathbb{R}}^{1\text{D top}}$ the topological 1D BV observables of the $\mathbb{R}$ -direction (Costello topological theory) and $\text{Obs}_{\mathbb{C}^2}^{4\text{D hol}}$ the holomorphic observables on $\mathbb{C}^2$ in the Dolbeault model. The factorisation-algebra product on $\mathbb{R} \times \mathbb{C}^2$ is the tensor product of the 1D E_1 -algebra product along $\mathbb{R}$ and the 4D holomorphic product on $\mathbb{C}^2$; in Dunn–Lurie form the ambient E_3 -structure factors as $E_1(\mathbb{R}) \otimes E_2(\mathbb{C}^2 \text{ in hol})$, with the second factor a holomorphic E_2 in the sense of Costello–Gwilliam Lemma §6.5.1. At the corner junction $\{0\} \times \mathbb{C} \subset \mathbb{R} \times \mathbb{C}^2$ the 1D topological factor degenerates to the point and the restricted factorisation algebra on $\mathbb{C}$ is the vertex algebra V_ψ ; the corner algebra $Y_{0,1,1}[\psi]$ is the image of this restriction under the three-way junction gluing.

Gaiotto–Rapčák [arXiv:1703.00982](#) corner vertex algebra, evaluated at $(N_1, N_2, N_3) = (0, 1, 1)$ and $\psi = -\epsilon_1/\epsilon_3$ (Li–Yamazaki [arXiv:2003.08909](#) §8.3.6 synthesis dictionary), is

$$Y_{0,1,1}[\psi] \supseteq^{\text{conj}} V(\widehat{\mathfrak{gl}}(1|1))_\psi^{qv},$$

a super-vertex algebra arising as the junction restriction of the Proposition 8.2 factorisation algebra with the $(0, 1, 1)$ gauge content on the three half-spaces. The Creutzig–Linshaw [arXiv:1706.00031](#) super- $\mathcal{W}$ -algebra $\mathcal{W}(\mathfrak{gl}(1|1))_\psi$ is a *logarithmic extension* of V_ψ^{qv} built from the isotropic odd simple roots (Creutzig–Ridout [arXiv:1303.0847](#); Ridout–Wood [arXiv:1203.3247](#)); the three objects V_ψ^{qv} , $Y_{0,1,1}[\psi]$, $\mathcal{W}(\mathfrak{gl}(1|1))_\psi$ are *genuinely distinct* at non-generic ψ , and the chain of identifications is not

a Zamolodchikov $\mathcal{W}$ -algebra in the classical Hamiltonian-reduction sense (the isotropic root system has no non-trivial Drinfeld–Sokolov reduction).

The central charge is *not* uniformly $c = 0$: along the family $\mathcal{W}(\mathfrak{gl}(1|1))_\psi$ the central charge $c(\psi)$ varies rationally in ψ ; $c = 0$ holds only on the self-dual $\epsilon_1 + \epsilon_2 = 0$ locus, where the super-trace annihilates the level contribution. At other equivariant chambers $c(\psi)$ is a nontrivial rational function of ψ (Creutzig–Gaiotto [arXiv:1706.02834](#) §5). From the Costello 5D hCS side, $c(\psi)$ is the one-loop wave-function renormalisation of the vertex-algebra current sector under the Costello–Li [arXiv:1606.00365](#) renormalisation scheme: the propagator $P_{L < \epsilon}$ on $\mathbb{R} \times \mathbb{C}^2$, regularised at length scale $L \in (0, \epsilon)$, yields a one-loop Feynman integral whose ψ -dependence is rational through the Costello–Yagi [arXiv:1810.01970](#) vertex computation on $\mathbb{C}^3$, specialised to the conifold half-space interface. The non-self-dual OPE family is **[Conjectural]** at all orders because super-Kontsevich–Tamarkin formality is E_2 -only for the $\mathfrak{gl}(1|1)$ super-Lie input (Ginzburg–Schedler [arXiv:0807.0174](#)); 2-loop and higher contributions to the OPE are not controlled by existing formality theorems (cf. Theorem 11.3). On the self-dual locus the OPE presentation reads

$$\begin{aligned} H(z)N(w) &\sim \frac{k}{(z-w)^2}, & \psi^+(z)\psi^-(w) &\sim \frac{k}{(z-w)^2} + \frac{H(w)}{z-w}, \\ H(z)H(w) &\sim 0, & N(z)N(w) &\sim 0 \quad (\text{Cartan self-isotropic}), \\ \psi^\pm(z)\psi^\pm(w) &\sim 0 \quad (\text{odd isotropic}), & N(z)\psi^\pm(w) &\sim \pm \frac{\psi^\pm(w)}{z-w}, \\ H(z)\psi^\pm(w) &\sim 0. \end{aligned}$$

The evaluation map $\text{ev}_\psi : Y^+(\widehat{\mathfrak{gl}(1|1)}) \rightarrow V_\psi^{qv}$ is a homomorphism of algebras on the *positive half* (not the full Yangian, and not an isomorphism); its image is the Gaiotto–Rapčák corner positive half. This map is the super avatar of the Costello–Yagi [arXiv:1810.01970](#) 5D-hCS-to-Yangian-VOA homomorphism (extended to the three-author setting in Costello–Gaiotto–Yagi [arXiv:2103.01835](#)),

$$\text{ev}_\psi^{\text{CGY}} : Y^+(\widehat{\mathfrak{g}}) \rightarrow V_\psi^{qv}[\mathfrak{g}] = \text{Obs}_{\mathfrak{g}}^{\text{5D hCS}}|_{\{0\} \times \mathbb{C}},$$

sourced at the corner junction of the Proposition 8.2 holomorphic-topological factorisation. For $\mathfrak{g}$ simply-laced with non-degenerate even Killing form (types A, D, E), CGY prove $\text{ev}_\psi^{\text{CGY}}$ is an *all-orders-in- $\hbar$* algebra homomorphism, matched on the CoHA side by Rapčák–Soibelman–Yang–Zhao [arXiv:2007.13365](#) Theorem B through $\text{CoHA}(\mathbb{C}_{\mathfrak{g}}^3) \twoheadrightarrow Y^+(\widehat{\mathfrak{g}})$. For $\mathfrak{g} = \mathfrak{gl}(1|1)$ with degenerate super-Killing form, the evaluation homomorphism is controlled only at tree and one-loop order (the wave-function sector); higher-loop orders are **[Conjectural]** pending super-Kontsevich–Tamarkin E_3 -formality for $\mathfrak{gl}(1|1)$ (Theorem 11.3). The full affine super-Yangian $Y(\widehat{\mathfrak{gl}(1|1)}) = Y^+ \bowtie Y^-$ acts on the field–state pairing module $V_\psi^{qv} \otimes (V_\psi^{qv})^\vee$ via the Drinfeld-double lift, *not* on V_ψ^{qv} itself as a single module — the standard Yangian-versus-evaluation-module distinction (Tsybaliuk [arXiv:1404.5240](#) §3). Writing “ $Y(\widehat{\mathfrak{gl}(1|1)}) \rightarrow V_\psi^{qv}$ ” is an abbreviation whose precise content is a Y^+ -action; the Y^- -action lives on the dual module.

Remark 8.3 (BV-anomaly first-principles status for $\mathfrak{gl}(1|1)$ 5D hCS). The one-loop BV anomaly of 5D hCS on $\mathbb{R} \times \mathbb{C}^2$ with gauge algebra $\mathfrak{g}$ is controlled by the totally-symmetric cubic invariant $d^{abc} \text{str}(t^a \{t^b, t^c\})$ on $\mathfrak{g}$ (Costello [arXiv:1303.2632](#) §3; Costello–Gwilliam [arXiv:2210.13036](#) §12 for the local-BV-cohomology framework; the obstruction lives in $H_{\text{loc}}^1(\mathfrak{g}, \mathcal{O}_{\text{loc}})$). For $\mathfrak{g} = \mathfrak{sl}_N$ with $N \geq 3$ this d^{abc} is nonzero and the theory is obstructed; for $\mathfrak{g} = \mathfrak{sl}_2$ or $\mathfrak{g}$ rank-one simply-laced the obstruction vanishes and all-orders quantisation proceeds (Costello–Gaiotto–Yagi [arXiv:1810.01970](#)). For the super case $\mathfrak{g} = \mathfrak{gl}(1|1)$ the supertrace satisfies $\text{str}(t^a \{t^b, t^c\}) = 0$ by direct basis enumeration (the N -Cartan has $\text{str} = 0$; the odd pair $\{\psi^+, \psi^-\}$ contributes with opposite signs), so the *one-loop wheel anomaly vanishes at leading order*. 2-loop and higher BV obstructions are not auto-killed: super-Kontsevich–Tamarkin formality for $\mathfrak{gl}(1|1)$ stops at E_2 (Ginzburg–Schedler [arXiv:0807.0174](#)), whereas all-orders quantisation of the generic- ψ family V_ψ^{qv} would require E_3 -formality of the classical

$\mathfrak{gl}(1|1)$ Poisson structure on $PV^\bullet(\mathbb{C}^3)$. The all-orders status of ev_ψ and of $Y_{0,1,1}[\psi]$ for generic ψ is therefore **[Conjectural]** (Theorem 11.3).

9. THE HOPF PAIRING AND THE DRINFELD DOUBLE

The CoHA $Y^+ = \text{CoHA}(\mathbf{Y})$ carries, by Davison [arXiv:1311.6989](#) Thm. A, a natural bialgebra structure: cohomological Hall product $m : Y^+ \otimes Y^+ \rightarrow Y^+$ and Davison coproduct $\Delta : Y^+ \rightarrow Y^+ \otimes Y^+$ (direct-sum pullback plus specialisation to the coherent-sheaf moduli stratum). The nil-radical positive half $Y^+ \subset Y(\widehat{\mathfrak{gl}(1|1)})^{\text{con}}$ is, in Drinfeld-new presentation, generated by $\{e^{(a)}(u) : a \in \{0, 1\}\}$; dually Y^- by $\{f^{(a)}(u)\}$. The question is whether there is a canonical Hopf pairing $\langle \cdot, \cdot \rangle : Y^+ \otimes Y^- \rightarrow \mathbb{C}((\hbar))$ making $D(Y^+)$ a Drinfeld double, and whether that pairing is non-degenerate. The section proceeds through five attack-heal cycles: (1) universal R -matrix intertwining produces the pairing; (2) triangular decomposition forces non-degeneracy; (3) the super sign $c_{11} = -1$ descends from the Kulish–Sklyanin residue at $u = 0$; (4) restriction to the even diagonal cross-checks against Tsybaliuk / Schiffmann–Vasserot; (5) the Drinfeld double recovers the full affine super-Yangian.

9.1. Cycle 1 — Universal R -matrix intertwining $Y^+ \otimes Y^-$.

Proposition 9.1 (Universal R -matrix determines a Hopf pairing). **[Proved elsewhere]** (Drinfeld 1987 ICM §8; Etingof–Kazhdan [arXiv:q-alg/9510022](#) Prop. 2.1 and §6.1 for the super extension.) *For any quasi-triangular Hopf algebra (H, Δ, R) admitting a triangular decomposition $H = H^- \otimes H^0 \otimes H^+$, the universal R -matrix factorises uniquely as $R = R^+ \cdot R^0 \cdot R^-$ with $R^\pm \in H^\pm \hat{\otimes} H^\mp$ and $R^0 \in H^0 \hat{\otimes} H^0$. The factor R^+ specifies a canonical bilinear Hopf pairing $\langle \cdot, \cdot \rangle : H^+ \otimes H^- \rightarrow \mathbb{C}[[\hbar]]$ by*

$$R^+ = \sum_{\alpha} e_{\alpha} \otimes f^{\alpha}, \quad \langle e_{\alpha}, f^{\beta} \rangle = \delta_{\alpha}^{\beta},$$

satisfying $\langle ab, x \rangle = \langle a \otimes b, \Delta(x) \rangle$ and $\langle a, xy \rangle = \langle \Delta(a), x \otimes y \rangle$.

Derivation sketch. Apply Drinfeld 1987 §8 to $Y(\widehat{\mathfrak{gl}(1|1)})^{\text{con}}$, whose universal R -matrix exists by Kulish–Sklyanin 1982 at rational level. The triangular decomposition $Y = Y^- \otimes Y^0 \otimes Y^+$ is the super Gauss factorisation $T(u) = F(u)D(u)E(u)$ of §7 (Zhang [arXiv:q-alg/9701011](#) Thm. 2.3). Etingof–Kazhdan [arXiv:q-alg/9510022](#) §6.1 extends the factorisation apparatus to the super setting: R^+ is the generating series of a dual basis, so the pairing is canonical once R is canonical. $\square$

Remark 9.2 (Existence vs. explicit formula). Proposition 9.1 shows the pairing *exists* and is unique up to triangular-decomposition convention; it does not produce an explicit formula for $\langle e_m^{(a)}, f_n^{(b)} \rangle$ on Drinfeld-new modes. That requires pushing the universal R -matrix through the RTT-to-new transition of §7 (Ding–Frenkel [arXiv:q-alg/9608002](#) super-version; Zhang [arXiv:q-alg/9701011](#)) and reading off the mode expansion, executed in Cycle 9.3.

9.2. Cycle 2 — Non-degeneracy and the Drinfeld double.

Proposition 9.3 (Non-degeneracy of the Hopf pairing at rational level). **[Proved elsewhere]** (Drinfeld 1987 §13; Gow [arXiv:math/0411166](#) Thm. 1.1 super; Nazarov 1991 *Lett. Math. Phys.* 21.) *The Hopf pairing $\langle \cdot, \cdot \rangle$ of Proposition 9.1 is non-degenerate on each finite-weight component $Y_{\gamma}^+ \otimes Y_{-\gamma}^-$ ($\gamma \in \mathbb{Z}_{\geq 0}^2$ the (d_0, d_1) -charge lattice); the induced map $Y_{\gamma}^+ \rightarrow (Y_{-\gamma}^-)^*$ is a $\mathbb{C}[[\hbar]]$ -linear isomorphism after $\hbar$ -adic completion.*

Derivation sketch. Three independent routes converge. **(a)** Drinfeld 1987 §13: the kernel $K_{\gamma} \subset Y_{\gamma}^+$ (were it nontrivial) is a coideal, $\Delta K \subset K \otimes Y^+ + Y^+ \otimes K$; the coproduct on Drinfeld-new generators is explicit (Davison [arXiv:1311.6989](#) Thm. A), and $K_{\gamma} = 0$ follows by induction on $|\gamma|$. **(b)** Gow [arXiv:math/0411166](#) Thm. 1.1 provides a PBW basis $\{e_{ij}^{(r)}\}_{i < j, r \geq 0}$ of Y^+ in RTT presentation

with matching dual basis in Y^- , making non-degeneracy visible by matrix inversion. **(c)** Nazarov 1991 Thm. 2 computes $\langle t_{ij}^{(r)}, t_{kl}^{(s)} \rangle = (-1)^{|i||k|+|j||l|} \delta_{il} \delta_{jk} [u^{-r} v^{-s}] \log(u - v + \hbar)$, whose log-kernel is non-degenerate on each homogeneous component. $\square$

Corollary 9.4 (Drinfeld double is well-defined). **[Proved elsewhere]** (Drinfeld 1987 §13; Majid 1995 *Foundations of Quantum Group Theory* §7.1.1.) *Non-degeneracy (Proposition 9.3) makes $D(Y^+) = (Y^+ \otimes (Y^-)^{\text{op}}) / \langle \text{pairing} \rangle$ a well-defined Hopf algebra with Drinfeld commutation*

$$e f = \sum \langle e_{(1)}, S(f_{(2)}) \rangle \langle e_{(3)}, f_{(1)} \rangle f_{(2)} e_{(2)}$$

(Sweedler notation), producing $Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ on Drinfeld-new level.

9.3. Cycle 3 — The super sign $c_{11} = -1$ from the R -matrix residue.

Lemma 9.5 (Super sign from the Kulish–Sklyanin residue at $u = 0$). **[Proposition]** *Let $R(u)$ be the Kulish–Sklyanin super- R -matrix on $\mathbb{C}^{1|1} \otimes \mathbb{C}^{1|1}$ from (9). In the $(2, 2; 2, 2)$ -block (both tensor factors odd)*

$$R_{22;22}(u) = 1 - \frac{\hbar}{u}, \quad \text{Res}_{u=0} [R_{22;22}(u) - 1] = -\hbar.$$

The minus sign is forced by $P^{\text{super}}(e_2 \otimes e_2) = (-1)^{|e_2||e_2|} e_2 \otimes e_2 = -(e_2 \otimes e_2)$, not by convention.

Proof. Direct from (9): the $(2, 2; 2, 2)$ entry is $1 - \hbar/u$; its residue at $u = 0$ of $R - 1$ is $-\hbar$. The sign traces to the super tensor-flip $P^{\text{super}} = \sum_{i,j} (-1)^{|i||j|} E_{ij} \otimes E_{ji}$, evaluating on $e_2 \otimes e_2$ as $-e_2 \otimes e_2$. The $-\hbar$ residue is a theorem of the $\mathbb{Z}/2$ -grading, not a normalisation choice. $\square$

Proposition 9.6 (Super sign propagates to the Drinfeld-new pairing). **[Conjectural]** *Under the RTT-to-new Gauss decomposition of §7, the $-\hbar$ residue of Lemma 9.5 propagates to a sign $(-1)^a$ on the Drinfeld-new pairing:*

$$\text{Res}_{u=v} \langle e^{(a)}(u), f^{(a)}(v) \rangle = (-1)^a \frac{\hbar}{u-v} + O(\hbar^2).$$

Derivation and honest scope. At leading order the RTT pairing $\langle t_{ij}^{(1)}, t_{kl}^{(1)} \rangle$ inherits the Nazarov 1991 sign $(-1)^{|i||k|+|j||l|}$. Under Gauss $T = FDE$, the Drinfeld-new current $e^{(a)}(u)$ identifies with $t_{a,a+1}(u)$ up to D -correction, $f^{(a)}(u)$ with $t_{a+1,a}(u)$. Substituting $(i, j, k, l) = (a, a+1, a+1, a)$:

$$(-1)^{|a||a+1|+|a+1||a|} = \begin{cases} (-1)^{0 \cdot 1 + 1} = -1 & a = 0, \\ (-1)^{1 \cdot 0 + 0} = +1 & a = 1, \end{cases}$$

giving $(-1)^{a+1}$, opposite to the desired $(-1)^a$. The discrepancy is absorbed by the antipode sign in the $F(u)$ -factor (Zhang [arXiv:q-alg/9701011](#) §3; Gow [arXiv:math/0411166](#) Rem. 2.4): the antipode of $F_{a+1,a}(u)$ carries an extra -1 on the odd sector, combining with the Nazarov sign to produce $(-1)^a$ overall. Fully rigorous propagation to all mode orders (m, n) is not carried out here; hence **[Conjectural]** at the explicit-mode-formula level, while Lemma 9.5 itself is theorem. $\square$

9.4. Cycle 4 — Even-only cross-check against $Y^+(\widehat{\mathfrak{gl}}_1)$.

Proposition 9.7 (Even diagonal matches Tsymbaliuk / Schiffmann–Vasserot). **[Proved elsewhere]** (Tsymbaliuk [arXiv:1404.5240](#) Prop. 4.3; Schiffmann–Vasserot [arXiv:1202.2756](#) Thm. 1.2.) *Restrict the conjectural pairing of Conjecture 9.10 below to $a = b = 0$. Then*

$$\langle e_m^{(0)}, f_n^{(0)} \rangle_{\hbar} = \hbar^{-1} \frac{\binom{m+n}{m}}{m+n+1},$$

matching Tsymbaliuk [arXiv:1404.5240](#) Prop. 4.3 for $Y^+(\widehat{\mathfrak{gl}}_1)$ under Schiffmann–Vasserot normalisation $\hbar = h_1 h_2 / h_3$, equivalently the Feigin–Odesskii shuffle inner-product on $\mathbb{C}^\times$ -equivariant cohomology of Hilbert schemes on $\mathbb{C}^2$.

Derivation. The even generator $e^{(0)}(u) = \sum_m e_m^{(0)} u^{-m-1}$ corresponds under the Schiffmann–Vasserot dictionary to multiplication by u^m on $H_T^*(\text{Hilb}^n(\mathbb{C}^2))$; Tsymbaliuk [arXiv:1404.5240](#) Prop. 4.3 gives the dual-basis pairing $\binom{m+n}{m}/(m+n+1) \cdot (h_3/h_1 h_2)$. Multiplied by $\hbar^{-1}$ with $\hbar = h_1 h_2 / h_3$ this is $\binom{m+n}{m}/(m+n+1)$. No super sign enters: the $a = b = 0$ sector is bosonic; the conifold even half restricts to Nakajima’s $H^*(\text{Hilb}(\mathbb{C}^2))$ via the chart projection $U_{\pm} = \mathbb{C}^3 \supset \mathbb{C}^2$. This is the chamber-independent statement (Theorem 10.2) on the even sublattice. $\square$

Remark 9.8 (Why the odd diagonal is harder). The odd diagonal $a = b = 1$ admits no commutative Hilbert-scheme model. It is accessed instead via the Davison–Hennecart–Schlegel-Mejia [arXiv:2303.12592](#) critical CoHA, where signs come from the vanishing-cycle sheaf $\varphi_{W_{\text{con}}}$ and the quartic potential. Their Thm. 6.1 establishes a bialgebra isomorphism of odd generators with a super-Yangian positive half; the explicit c_{11} coefficient is not computed there. This gap is why Conjecture 9.10 below remains **[Conjectural]**.

9.5. Cycle 5 — Drinfeld double $D(Y^+) = Y(\widehat{\mathfrak{gl}}(1|1))$.

Theorem 9.9 (Drinfeld-double recovers the full super-Yangian). **[Proved elsewhere]** (Ding–Frenkel [arXiv:q-alg/9608002](#) for $Y(\widehat{\mathfrak{gl}}_n)$; Zhang [arXiv:q-alg/9701011](#) §3 super; Gow [arXiv:math/0411166](#) Thm. 1.1 PBW.) *Non-degeneracy (Proposition 9.3) and the Gauss decomposition $T(u) = F(u)D(u)E(u)$ of Route R4 together imply that the Drinfeld double of Y^+ with respect to the Hopf pairing of Corollary 9.4 recovers the full affine super-Yangian:*

$$D(Y^+) = \langle Y^+ \otimes Y^-, \text{Drinfeld commutation} \rangle \simeq Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}.$$

Derivation sketch. Gauss decomposition identifies $E(u)$ with Y^+ (upper-unipotent), $F(u)$ with Y^- (lower-unipotent), $D(u)$ with Y^0 (diagonal). The RTT relation $R(u-v)T_1(u)T_2(v) = T_2(v)T_1(u)R(u-v)$ decomposes after Gauss substitution into (a) within- E relations giving Y^+ structure, (b) within- F relations giving Y^- structure, (c) cross-terms $E \otimes F$ giving Drinfeld commutation matching Corollary 9.4. This is Zhang [arXiv:q-alg/9701011](#) Thm. 3.1 for $DY(\widehat{\mathfrak{gl}}(1|1))$ at toroidal level; Galakhov–Li–Yamazaki [arXiv:2106.01230](#) establishes the conifold-specific toroidalisation, giving $DY(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$. Recovery is theorem, independent of the conjectural c_{11} subleading correction. $\square$

9.6. Synthesis: conjectural mode formula and honest status.

Conjecture 9.10 (Hopf pairing explicit formula). **[Conjectural]** (S5 per AP-CY297.) *Assuming the Gauss-decomposition sign propagation of Proposition 9.6 extends to all mode orders (m, n) , the Drinfeld-new Hopf pairing $\langle \cdot, \cdot \rangle_{\hbar} : Y^+ \otimes Y^- \rightarrow \mathbb{C}((\hbar))$ takes the explicit form*

$$(10) \quad \langle e_m^{(a)}, f_n^{(b)} \rangle_{\hbar} = \delta^{a,b} (-1)^a \hbar^{-1} \frac{\binom{m+n}{m}}{m+n+1} (1 + \hbar c_{ab} \delta_{m,n}),$$

with $c_{00} = 0$, $|c_{11}| = 1$ forced by Lemma 9.5, and sign of c_{11} determined by the antipode-in-coproduct convention:

- Nazarov 1991 / Kulish–Sklyanin 1982 convention (*antipode S post-multiplies*): $c_{11} = -1$, consistent with the $-\hbar$ residue of Lemma 9.5;
- Gow 2005 [arXiv:math/0411166](#) / Peng 2011 convention (*S pre-multiplies*): $c_{11} = +1$.

Off-diagonal entries ($a \neq b$) vanish by parity orthogonality. The $\hbar$ -deformation parameter relates to the equivariant parameters $h_1, h_2, h_3 = -h_1 - h_2$ of $T \subset \text{GL}(3)$ by the Schiffmann–Vasserot normalisation $\hbar = h_1 h_2 / h_3$ ([arXiv:1202.2756](#) §3.2). The even diagonal ($a = 0$) is theorem (Proposition 9.7 via Tsymbaliuk [arXiv:1404.5240](#) Prop. 4.3 and Schiffmann–Vasserot [arXiv:1202.2756](#) Thm. 1.2); the odd-diagonal ($a = 1$) mode-level formula is conjectural, pending a direct Gauss-decomposition residue computation verifying $\binom{m+n}{m}/(m+n+1)$ against the Nazarov 1991 Thm. 2 Berezinian-residue form

$$\langle t_{ij}^{(r)}, t_{kl}^{(s)} \rangle = (-1)^{|i||k|+|j|} \delta_{il} \delta_{jk} [u^{-r} v^{-s}] [\log \text{Ber}_q T(u) \cdot k^{\text{off}}(u-v)],$$

with offset kernel $k^{\text{off}}(u) = (u + \varepsilon_1)(u + \varepsilon_2)/[u(u + \varepsilon_1 + \varepsilon_2)]$. The binomial form is a legitimate shadow under the Drinfeld–Jimbo evaluation map to the bosonic subalgebra $Y^+(\widehat{\mathfrak{gl}}_1)$ (Cycle 9.4); its full validity on the odd diagonal requires verification against Nazarov’s log-kernel and an independent Gow–Peng cross-derivation ([arXiv:math/0411166](#); Peng 2011 J. Algebra 325; Zhang [arXiv:q-alg/9701011](#)), open at mode level.

Remark 9.11 (Three independent cross-checks for the leading term). Three parallel derivations converge on the leading binomial coefficient. **(a)** Maulik–Okounkov stable-envelope residue

$$\text{Res}_{u=0} R^{MO}(u) = \frac{h_1 h_2}{h_1 + h_2} \cdot P^{\text{super}},$$

matches (10) at $(m, n) = (0, 0)$ under the Costello–Gaiotto normalisation $h_1 = \sqrt{\hbar}$ (Maulik–Okounkov [arXiv:1211.1287](#) §4.5). **(b)** Soibelman–Gelfand refined Ext-pairing $\mathbf{E}_h(M, N) = \sum_i (-\hbar^{1/2})^i \dim \text{Ext}^i(M, N)$ with Milnor–Moore extension matches the leading Drinfeld–double pairing coefficient. **(c)** The RTT Hopf pairing of Nazarov 1991 Thm. 2 and Gow [arXiv:math/0411166](#) Thm. 1.1 agrees under the Ding–Frenkel [arXiv:q-alg/9608002](#) / Zhang [arXiv:q-alg/9701011](#) RTT-to-new dictionary on the leading term. These three derivations agree on the binomial coefficient but neither independently establishes the $c_{11} = -1$ subleading correction at all modes (m, n) : (a) gives only the residue at $u = 0$; (b) gives only the Ext-pairing leading term; (c) produces a log-kernel rather than a pure binomial. The odd-diagonal pairing is therefore **[Conjectural]** at subleading order; the even diagonal is theorem (Proposition 9.7).

Theorem 9.12 (Hopf status of the Drinfeld double). **[Proved elsewhere]** (Etingof–Kazhdan [arXiv:q-alg/9510022](#) and its super extension in [arXiv:q-alg/9510023--25](#); Jimbo–Konno 1997 for the elliptic case; Zhang [arXiv:q-alg/9701011](#) for the super-Yangian Gauss decomposition.) *The Drinfeld double $D(Y^+) = Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ of Theorem 9.9 is*

- strict Hopf at rational, trigonometric, and toroidal levels (associator $\Phi = 1$);
- quasi-Hopf at elliptic level, with associator Φ_{ell} built from the Felder–Jimbo–Konno dynamical R -matrix and theta-function boundary data.

The “quasi-Hopf from super + Serre” intuition fails: $\mathfrak{gl}(1|1)$ ’s supersymmetric pairing is even-split, so the Etingof–Kazhdan obstruction $H_{\text{EK}}^3(\mathfrak{gl}(1|1), \mathfrak{gl}(1|1)^{\otimes 3}) = 0$; KW Serre relations are wheel conditions on the shuffle kernel, not coproduct associators; the quartic potential W_{con} enters the Jacobi relations, not the coassociator. Hopf status is independent of Conjecture 9.10’s subleading c_{11} correction: strict Hopf at rational level follows from non-degeneracy (Proposition 9.3) alone.

10. CHAMBER STRUCTURE

Theorem 10.1 (Chamber structure, Nagao–Nakajima [arXiv:0809.2992](#)). **[Proved elsewhere]** *The stability space of $J(Q_{\text{con}}, W_{\text{con}})$ -modules forms a $W(\widehat{\mathfrak{sl}}_2)$ -torsor with infinitely many chambers generated by Seiberg dualities (equivalently, Keller–Yang [arXiv:0906.0761](#) quiver mutations) acting as simple reflections on the charge lattice. Named fundamental chambers:*

- Non-commutative (Szendrői): $(s_0, s_1) = (0, 0)$,
- Donaldson–Thomas (large-volume): $(s_0, s_1) = (0, -1)$ or $(-1, 0)$,
- Pandharipande–Thomas (stable pairs): $(s_0, s_1) = (-1, -1)$,
- Empty: $(s_0, s_1) = (+\infty, +\infty)$ limit.

Shifted quiver Yangian (Galakhov–Li–Yamazaki [arXiv:2106.01230](#)): each chamber corresponds to a shift $\mathbf{s} \in \mathbb{Z}^2$ with Braverman–Finkelberg–Nakajima shift homomorphism $\iota_{\mathbf{s}, \mathbf{s}+1} : Y_{\mathbf{s}} \hookrightarrow Y_{\mathbf{s}+1}$. The underlying algebra $Y^+(\widehat{\mathfrak{gl}}(1|1))$ is chamber-independent; representations are chamber-parametrised.

Theorem 10.2 (Chamber-independence of the Hopf pairing). **[Proved elsewhere]** (Joyce–Song [arXiv:0810.5645](#) Thm. 5.11 + KS wall-crossing) *The Joyce–Song integration map $I : \text{SF}^{\text{ind}}(\mathfrak{M}) \rightarrow$*

$\widehat{U}(\mathfrak{g}_{\text{BPS}})[[\hbar]]$ is KS -equivariant, and wall-crossing corrections $\alpha_W = \sum_{\gamma=k\gamma_W} \Omega(\gamma)[\gamma]/k^2$ land in orthogonal K_0 -graded pieces. Hence the Hopf-pairing matrix entries are literally equal across chambers, not merely projectively equivariant.

11. PHYSICAL ORIGIN — THE M-THEORY PARENT

Conjecture 11.1 (M-theory parent of the conifold super-Yangian). **[Conjectural]** *The M-theory parent of $Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ is the 6D (2,0) A_{K-1} theory on $\mathbb{R}^{1,3} \times T^2$ with K M5-branes wrapped on $\mathbb{P}^1 \subset \mathbf{Y}$, normal-bundle twist $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$, Ω -deformation on transverse Taub–NUT, and Costello–Gwilliam-twist on the CY_3 target. Dimensional reductions:*

- Bulk: 5D holomorphic Chern–Simons on $\mathbb{R} \times \mathbf{Y}$ with gauge $\mathfrak{gl}(1|1)$ -valued (open per Theorem 11.3).
- Boundary: 3D $\mathcal{N} = 4$ sigma model to $\text{Hilb}^n(\mathbf{Y})$ whose Coulomb-branch twist is conjecturally the super-Yangian positive half $Y^+(\widehat{\mathfrak{gl}}(1|1))$ at infinity.

The Coulomb-branch identification transfers Braverman–Finkelberg–Nakajima $\mathcal{A}_h(K3) = Y(\mathfrak{g}_{K3})$ ([arXiv:1601.03586](#), [arXiv:1604.03625](#)) and Nakajima [arXiv:1606.09041](#) to the conifold setting, with Putrov–Yagi [arXiv:1704.04255](#) and Kamnitzer–Pham–Weekes [arXiv:1806.07146](#) providing the 3D $\mathcal{N} = 4$ Coulomb-branch extension machinery for ADE quivers. The super-quiver Klebanov–Witten extension, however, is not proved rigorous at the Coulomb-branch level in any published source: BFN requires a non-degenerate Killing form, which $\mathfrak{gl}(1|1)$ lacks, and the Putrov–Yagi 4D/3D descent was established for $\mathcal{N} = 2$ SCFTs whereas KW is $\mathcal{N} = 1$ (cf. Theorem 11.2). This statement is therefore a synthesis by analogy with the Costello–Paquette $\mathbb{C}^3$ construction ([arXiv:2001.02177](#) and [arXiv:2201.02595](#)), not an explicit theorem. The specific wrap on a compact $\mathbb{P}^1$ with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$, combined with the super gauge extension, lies beyond the Costello–Paquette theorems as written and beyond the BFN/Putrov–Yagi Coulomb-branch framework as it stands.

Theorem 11.2 (Superconformal BLLPRvR route is unavailable). **[Proved elsewhere]** (by type-mismatch) *The Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees [arXiv:1312.5344](#) 4D/2D chiral-algebra correspondence applies to 4D $\mathcal{N} = 2$ superconformal theories. The 4D field theory on an M5-brane wrapped on $\mathbb{P}^1 \subset \mathbf{Y}$ with KW normal bundle is the Klebanov–Witten $\mathcal{N} = 1$ SCFT, not $\mathcal{N} = 2$. Hence BLLPRvR does not directly apply to this boundary theory. The identification $\text{CoHA}(\mathbf{Y}) = Y^+(\widehat{\mathfrak{gl}}(1|1))$ therefore runs through the Rapčák–Soibelman–Yang–Zhao [arXiv:1810.10402](#) route on the CoHA side, not through BLLPRvR.*

Theorem 11.3 (5D hCS super extension is open). **[Open]** *The Costello–Gaiotto–Yagi all-orders theorem for $5D \text{ hCS} \rightarrow \text{Yangian VOA}$ (Costello 2013 [arXiv:1303.2632](#); Costello–Yagi [arXiv:1810.01970](#); the three-author CGY paper with Gaiotto is [arXiv:2103.01835](#)) requires simply-laced gauge algebra with non-degenerate even Killing form. The super case $\widehat{\mathfrak{gl}}(1|1)$ has degenerate Killing form (the supertrace annihilates K); super Kontsevich–Tamarkin formality is E_2 -only (Ginzburg–Schedler [arXiv:0807.0174](#)), not E_3 ; higher-loop obstructions in $H_{\text{loc}}^1(\mathfrak{gl}(1|1), \mathcal{O}_{\text{loc}})$ are not auto-killed. One-loop wheel vanishing $d_{\text{super}}^{abc} = 0$ for $\mathfrak{gl}(1|1)$ is verified by direct basis computation; 2-loop and higher remain open.*

Remark 11.4 (Per-chart $\widehat{\mathfrak{gl}}_1$ is all-orders rigorous). The per-chart abelian 5D hCS with $\widehat{\mathfrak{gl}}_1$ gauge on each $U_{\pm} \cong \mathbb{C}^3$ is all-orders rigorous (Costello–Li, Schiffmann–Vasserot); the difficulty is in the super-assembly of two such algebras via the FM transition.

Remark 11.5 (Anomaly vanishing on $\mathbf{Y}$). The BCOV 1-loop anomaly $\alpha_{\text{BCOV}} = (\chi/24)[\Omega]^{0,1} \in H^{0,1}(\mathbf{Y})$ (Costello–Li 2016) vanishes because $\mathbf{Y}$ is a rank-2 vector bundle over $\mathbb{P}^1$ (hence homotopy-equivalent to $\mathbb{P}^1$ via the affine bundle retraction), and $H^{0,1}(\mathbb{P}^1) = h^{0,1}(\mathbb{P}^1) = 0$. The statement “ $\mathbf{Y}$ is affine” common in earlier drafts is inaccurate — $\mathbf{Y}$ is quasi-projective, not affine, since $\mathbb{P}^1$ is proper; the correct statement is “ $\mathbf{Y}$ retracts to $\mathbb{P}^1$ and $H^{0,1}(\mathbb{P}^1) = 0$ ”.

Remark 11.6 (Non-abelian $K \geq 2$ obstructions). For A_{K-1} stacks of M5-branes, the 1-loop $d^{abc}d^{abc}$ anomaly obstructs perturbative consistency at $K \geq 3$. The $K = 2$ case is special because $\mathfrak{sl}_2$ has $d^{abc} = 0$ (Costello 2015 [arXiv:1112.0816](#) §4-5; CLAUDE.md key-fact). Note $\mathfrak{sl}_2 \neq \mathfrak{gl}_2$: the $\mathfrak{gl}_2 = \mathfrak{sl}_2 \oplus \mathfrak{u}(1)$ decomposition has vanishing symmetric cubic on $\mathfrak{sl}_2$ and BRST-trivial $\mathfrak{u}(1)$, so $K = 2$ gives a well-defined 5D CS at the gauge-algebra level.

12. SUPER-RIGIDITY, GOPAKUMAR–VAFA, AND REFINED TOPOLOGICAL VERTEX

M-theory on $\mathbf{Y} \times S^1$ with M2-branes wrapping $k[\mathbb{P}^1] \in H_2(\mathbf{Y}, \mathbb{Z})$ presents these wrapped branes as 5D BPS hypermultiplets charged under the graviphoton F_{RR} ; integrating them out by a Schwinger one-loop produces the $2 \sinh(kg_s/2)$ denominator and the Li_3 -polylogarithm envelope of the conifold free energy (Gopakumar–Vafa [arXiv:hep-th/9809187](#) Part I; [arXiv:hep-th/9812127](#) Part II). The two-parameter $(\varepsilon_1, \varepsilon_2)$ -refinement deforms the Schwinger integrand to a product of two sinh factors (Nekrasov–Okounkov [arXiv:hep-th/0306238](#) §7; geometric realisation Iqbal–Kozcaz–Vafa [arXiv:hep-th/0701156](#)).

Theorem 12.1 (Super-rigidity of $(-1, -1)$ -curves). **[Proved elsewhere]** (Bryan–Pandharipande [arXiv:math/0009025](#) Thm. 1.2) *A rigidly embedded $\mathbb{P}^1 \subset X$ in a smooth Calabi–Yau threefold X with normal bundle $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$ is super-rigid: it has no infinitesimal deformations as a map from a smooth curve of any genus to X . Hence Gromov–Witten contributions to all genera come only from multiple covers of the embedded $\mathbb{P}^1$.*

The multiple-cover Hodge-integral formula for super-rigid $\mathbb{P}^1$ (Faber–Pandharipande [arXiv:math/9810173](#) for the multiple-cover integrals; Pandharipande [arXiv:math/9902107](#) for degenerate contributions; Bryan–Pandharipande [arXiv:math/0412005](#) for the local Gromov–Witten theory of curves that assembles these integrals into the toric vertex) combined with the M-theoretic BPS-counting resummation of Gopakumar–Vafa [arXiv:hep-th/9809187](#) Part I eqs. (2.7)–(2.10) and Part II [arXiv:hep-th/9812127](#) for the higher-genus structure, yields the BPS per-genus ansatz

$$F^{\text{GV}}(\mathbf{Y}; \lambda, Q) = \sum_{g \geq 0} \sum_{\beta \in H_2(\mathbf{Y}, \mathbb{Z})} \sum_{k \geq 1} n_{\beta}^g \frac{1}{k} (2 \sinh(k\lambda/2))^{2g-2} Q^{k\beta},$$

where $\lambda = g_s$ is the string coupling, $Q^{\beta} = e^{-\int_{\beta} \omega}$ the Kähler-modulus character, and $n_{\beta}^g \in \mathbb{Z}_{\geq 0}$ the integer Gopakumar–Vafa invariants counting BPS states with (J_L^3) -spin g in charge class β . On the conifold the effective cone is generated by $\beta = [\mathbb{P}^1]$ and the only non-vanishing invariant is $n_{[\mathbb{P}^1]}^0 = 1$ with $n_{d[\mathbb{P}^1]}^g = 0$ for $(g, d) \neq (0, 1)$, which collapses the ansatz to the Schwinger-resummed form

$$F^{\text{GV}}(\mathbf{Y}; g_s, Q) = \sum_{k \geq 1} \frac{Q^k}{k (2 \sinh(kg_s/2))^2}.$$

The genus expansion $F^{\text{GV}} = \sum_{g \geq 0} g_s^{2g-2} F_g(Q)$ follows from the formal power-series identity $\frac{x^2}{(2 \sinh(x/2))^2} = \sum_{g \geq 0} \frac{(1-2g) B_{2g}}{(2g)!} x^{2g}$, giving

$$F_0(Q) = \text{Li}_3(Q), \quad F_g(Q) = \frac{(-1)^{g-1} B_{2g}}{2g(2g-2)!} \text{Li}_{3-2g}(Q) \quad (g \geq 2),$$

recorded in Mariño [arXiv:hep-th/0406005](#) eq. (2.68). The $g = 1$ contribution is the x^0 -term of $(2 \sinh(x/2))^{-2} = x^{-2} - \frac{1}{12} + \frac{x^2}{240} - \dots$ applied to $\text{Li}_1(Q) = -\log(1-Q)$, giving $F_1(Q) = -\frac{1}{12} \text{Li}_1(Q) = \frac{1}{12} \log(1-Q)$ (holomorphic-anomaly constant absorbed). The $(-1)^{g-1}$ prefactor multiplies the Bernoulli number B_{2g} , whose own sign is $(-1)^{g+1}$ for $g \geq 1$; the product is uniformly negative for $g \geq 2$, matching coefficient-extraction from the Schwinger form.

Theorem 12.2 (Szendrői NCDT partition function, two chambers distinguished). **[Proved elsewhere]** (Szendrői [arXiv:0705.3419](#) Thm. 2.6 for the chamber-specific NC result; Morrison–Mozgovoy–Nagao–Szendrői [arXiv:1107.5017](#) Thm. 2 for the two-variable refinement; Young [arXiv:0802.3948](#) for pyramid partitions.) *The curve-less Szendrői theorem gives the NCDT partition function indexed by pyramid partitions*

$$Z^{\text{NCDT}}(\mathbf{Y}; q) = \sum_{\pi \in \text{Pyr}} (-q)^{|\pi|} = M(-q)^2 \cdot \prod_{k \geq 1} (1 - (-q)^k)^{-2} \quad (\text{Szendrői chamber, single parameter}).$$

Passing to two variables (q, Q) adapted to the $D^b(\mathbf{Y})$ -tilt reorganisation, the full two-variable NCDT partition function is

$$Z^{\text{NCDT}}(\mathbf{Y}; q, Q) = M(-q)^2 \cdot \prod_{k \geq 1} (1 - Q(-q)^k)^k \prod_{k \geq 1} (1 - Q^{-1}(-q)^k)^k,$$

i.e. both Q and Q^{-1} factors are present in the Szendrői (non-commutative) chamber (MMNS [arXiv:1107.5017](#), eq. (1.1)). The DT (commutative) chamber partition function is the truncation

$$Z^{\text{DT}}(\mathbf{Y}; q, Q) = M(-q)^2 \prod_{k \geq 1} (1 - Q(-q)^k)^k,$$

obtained by absorbing the $\prod_k (1 - Q^{-1}(-q)^k)^k$ factor into the vacuum sector via a single Kontsevich–Soibelman wall-crossing operator $\mathcal{T}_{\gamma_\alpha}$ (Nagao–Nakajima [arXiv:0809.2992](#) for the conifold perverse-coherent wall-crossing framework; Nagao [arXiv:0809.2994](#) for the small-toric CY_3 chamber structure specialised to $\mathbf{Y}$; cross-reference [chapters/examples/coha_wall_crossing_platonic.tex](#) lines 990–995). Conflating the DT truncation with the full NC partition function is a common error; the Szendrői (NC) chamber carries both Q and Q^{-1} infinite products, whereas the DT chamber retains only the Q product.

Proposition 12.3 (Szendrői NCDT coincides with Bridgeland NCDT after wall-crossing). **[Proved elsewhere]** (Nagao–Nakajima [arXiv:0809.2992](#) for the categorical wall-crossing on the conifold; Bridgeland [arXiv:1002.4372](#) Thm. 6.3 for the motivic Hall-algebra framework; MMNS [arXiv:1107.5017](#) Thm. 2 for the two-variable match.) *The Szendrői NCDT generating series $Z^{\text{NCDT}}(\mathbf{Y}; q, Q)$, defined from pyramid partitions for the Jacobi algebra $J(Q_{\text{con}}, W_{\text{con}})$, equals the image of the Bridgeland NCDT generating series — the motivic Donaldson–Thomas invariant of the motivic Hall algebra of $D^b(J(Q_{\text{con}}, W_{\text{con}})\text{-mod})$ at the tilt-Noetherian heart $\mathcal{A}_{\text{NC}}$ — under the chain of Kontsevich–Soibelman wall-crossing automorphisms carrying $\mathcal{A}_{\text{NC}}$ to the large-volume heart $\text{Coh}(\mathbf{Y})$. The composite wall-crossing factor is $\prod_k (1 - Q^{-1}(-q)^k)^k$, contributing precisely the extra infinite product that distinguishes the Szendrői (NC) chamber from the DT (commutative) chamber, so that on the DT side the Szendrői and Bridgeland generating functions coincide as rational functions in (q, Q) .*

Proof. The categorical wall-crossing is the identity of hearts $\mathcal{A}_{\text{NC}} \rightarrow \text{Coh}(\mathbf{Y})$ along the Nagao–Nakajima perverse-coherent chamber sequence (one wall per Seiberg mutation s_0, s_1). Each wall-crossing automorphism is the KS operator $\mathcal{T}_\gamma = \exp(\sum_k \Omega(k\gamma)[k\gamma]/k^2)$ with $\Omega(\gamma)$ the refined DT invariant of the simple γ . Summing the generating series under these automorphisms and applying the Bridgeland rationality theorem [arXiv:1002.4372](#) Thm. 6.3 to $D^b(\mathbf{Y})$ converts Z^{NCDT} to Z^{DT} by absorbing precisely the $\prod_k (1 - Q^{-1}(-q)^k)^k$ factor; the identification on both sides is the content of MMNS [arXiv:1107.5017](#) Thm. 2. $\square$

Under $q = -e^{ig_s}$ the DT-chamber expression matches F^{GV} via the MNOP triangle of partition-function identities

$$\begin{array}{ccc} & Z^{\text{GW}}(\mathbf{Y}; g_s, Q) & \\ & \updownarrow^{\text{GW/DT}} & \\ Z^{\text{DT}}(\mathbf{Y}; q, Q) & \xleftarrow{\text{DT/PT}} & Z^{\text{PT}}(\mathbf{Y}; q, Q) \end{array}$$

with $q = -e^{ig_s}$ and a common reduced free energy F^{GV} on the vertices. The three edges are proved independently:

- *GW/DT (top edge)*: Maulik–Nekrasov–Okounkov–Pandharipande I [arXiv:math/0312059](#) Conj. 3R, extended in MNOP II [arXiv:math/0406092](#) to the equivariant and reduced theory; the toric-CY₃ case is proved for $\mathbf{Y}$ via the local-curve primitive Bryan–Pandharipande [arXiv:math/0412005](#) (the $(-1, -1)$ -cap computation that feeds the toric vertex), the full toric proof being completed in Maulik–Oblomkov–Okounkov–Pandharipande [arXiv:0809.3976](#). The 2005 Bryan–Pandharipande paper [arXiv:math/0412005](#) is the *local Gromov–Witten primitive* feeding MNOP; it is distinct from the 2001 super-rigidity primitive Bryan–Pandharipande [arXiv:math/0009025](#) of Theorem 12.1, and from the 1999 Hodge-integral input Pandharipande [arXiv:math/9902107](#) feeding the multiple-cover formula.
- *DT/PT (bottom edge)*: Pandharipande–Thomas [arXiv:0707.2348](#) define the stable-pair moduli; Bridgeland [arXiv:1002.4372](#) proves the DT/PT identity rationally via motivic Hall algebras; Toda [arXiv:0902.4371](#) gives the motivic DT/PT equivalence at refined level.
- *GW/PT (right edge)*: composite of the other two, consistent with the MNOP I rationality conjecture on toric $\mathbf{Y}$.

Remark 12.4 (Refined topological vertex: primary-source conventions). **[Convention-dependent]**

(Iqbal–Kozcaz–Vafa [arXiv:hep-th/0701156](#), eqs. (4.2)–(4.3); independent Awata–Kanno [arXiv:hep-th/0502061](#) derivation.) Iqbal–Kozcaz–Vafa present the refined $(-1, -1)$ -conifold partition function as

$$Z_{\text{top}}^{\text{ref}}(\mathbf{Y}; t, q, Q) = \prod_{i,j \geq 1} (1 - Q t^{i-1/2} q^{j-1/2}),$$

with the half-integer offsets $(i - 1/2, j - 1/2)$ a *convention choice* — Awata–Kanno’s equivalent form uses integer offsets $(i, j) \mapsto (i - 1, j - 1)$ with a prefactor $Q^{-1/2}$ absorbed into the Kähler modulus. The half-integer form is the one in IKV eq. (4.2) and is the form that makes the single-leg refined vertex $P_{\lambda\mu\nu}(q, t)$ Young-diagram-combinatorial. The sign convention q, t in IKV (§3) corresponds to the flop-invariant normalisation; the opposite-sign convention of Taki [arXiv:hep-th/0703133](#) differs by $(q, t) \mapsto (q^{-1}, t^{-1})$ on the unrefined locus $q = t$. The identification $t = e^{i\varepsilon_1}$, $q = e^{i\varepsilon_2}$ with Nekrasov’s Ω -background requires the following primary-source correspondence to be recorded: Nekrasov 2002 [hep-th/0206161](#) introduced the one-parameter Ω -background on $\mathbb{C}^2 \times S^1$; the *refined* two-parameter extension to $(\varepsilon_1, \varepsilon_2)$ for gauge theories was Nekrasov–Okounkov 2003 [hep-th/0306238](#), §7; the refined *topological vertex* was IKV [hep-th/0701156](#); the MNOP GW/DT correspondence [math/0312059](#) is unrefined. Any claim “Nekrasov partition function for conifold in Ω -background” must cite Nekrasov–Okounkov 2003 for the two-parameter refinement, not Nekrasov 2002 (one-parameter) and not MNOP (unrefined).

Remark 12.5 (Refined free energy: Nekrasov–Okounkov Schwinger form). **[Proved elsewhere]**

(Nekrasov–Okounkov [arXiv:hep-th/0306238](#) §7; IKV [arXiv:hep-th/0701156](#) §4; Nakajima–Yoshioka [arXiv:0905.0897](#) Thm. 1.1 for the K-theoretic gauge-theory / refined-DT bridge.) Taking the logarithm of the IKV product yields the refined conifold free energy

$$F^{\text{ref}}(\mathbf{Y}; \varepsilon_1, \varepsilon_2, Q) = - \sum_{i,j \geq 1} \log(1 - Q t^{i-1/2} q^{j-1/2}) = \sum_{k \geq 1} \frac{Q^k}{k (2 \sinh(k\varepsilon_1/2)) (2 \sinh(k\varepsilon_2/2))},$$

with $t = e^{i\varepsilon_1}$, $q = e^{i\varepsilon_2}$. Under the unrefined specialisation $\varepsilon_1 = \varepsilon_2 = g_s$ this collapses to $F^{\text{GV}}(\mathbf{Y}; g_s, Q) = \sum_{k \geq 1} Q^k / [k (2 \sinh(kg_s/2))^2]$, matching the Schwinger form of the preceding display. The two-sinh denominator records the Ω -deformed $\mathbb{C}_{\varepsilon_1, \varepsilon_2}^2$ equivariant Hilbert series at the origin. Nakajima–Yoshioka [arXiv:0905.0897](#) Thm. 1.1 proves the K-theoretic version for moduli of framed sheaves on $\mathbb{P}^2$, confirming the same two-parameter Schwinger structure on the gauge-theory side.

Conjecture 12.6 (Algebraic super-rigidity of $Y^+(\widehat{\mathfrak{gl}}(1|1))$ under $(\varepsilon_1, \varepsilon_2)$ -deformation). **[Conjectural]** (first stated here as a conjecture; unrefined $\widehat{\mathfrak{gl}}_1$ -cousin is the Schiffmann–Vasserot rigidity result [arXiv:1202.2756](#); the ε_+ -gauge mechanism traces to Nekrasov–Okounkov [arXiv:hep-th/0306238](#) §7; the 5D hCS super obstruction enters via the three-author Costello–Gaiotto–Yagi [arXiv:2103.01835](#), not the two-author Costello–Yagi [arXiv:1810.01970](#).) *The chain-level deformation of the positive half $Y^+(\widehat{\mathfrak{gl}}(1|1))$ by the two parameters $(\varepsilon_1, \varepsilon_2)$ of the refined Ω -background is algebraically super-rigid: the cohomology group $\mathrm{HH}_{\mathrm{Hopf}}^2(Y^+(\widehat{\mathfrak{gl}}(1|1)); Y^+(\widehat{\mathfrak{gl}}(1|1)))$ governing graded Hopf-superalgebra deformations is one-dimensional, spanned by the class of the refined Ω -background with first-order parameter $\varepsilon_1\varepsilon_2$. The sum $\varepsilon_+ := \varepsilon_1 + \varepsilon_2$ is absorbed into the Nekrasov–Okounkov ε_+ -rescaling gauge equivalence ([hep-th/0306238](#) §7, a one-parameter dilatation of the Drinfeld current generators targeting the supertrace-degenerate direction of the $\widehat{\mathfrak{gl}}(1|1)$ Killing form — the same degeneration that obstructs the 5D hCS super-extension in Theorem 11.3), singling out $\varepsilon_1\varepsilon_2$ as the unique physical first-order deformation and reproducing the Nekrasov–Okounkov refined Schwinger form of Remark 12.5 at the level of super-characters. Geometric super-rigidity (Theorem 12.1) is a distinct statement: a geometric fact about deformations of the embedded $\mathbb{P}^1$, not an algebraic fact about the BPS algebra. The two are linked only through the CoHA dictionary $\mathrm{CoHA}(\mathbf{Y}) \simeq Y^+(\widehat{\mathfrak{gl}}(1|1))^{\mathrm{con}}$.*

Remark 12.7 (Why the unrefined rigidity is Schiffmann–Vasserot and the refined is open). Schiffmann–Vasserot [arXiv:1202.2756](#) establish algebraic rigidity of $Y^+(\widehat{\mathfrak{gl}}_1)$ under the single-parameter $\hbar$ -deformation in the abelian (even) case. The super case under the two-parameter $(\varepsilon_1, \varepsilon_2)$ deformation is not covered: (i) Ginzburg–Schedler super formality [arXiv:0807.0174](#) terminates at E_2 rather than E_3 , removing the Kontsevich–Tamarkin formality input that drives the SV rigidity proof; (ii) the super Hopf-algebra cohomology $\mathrm{HH}_{\mathrm{Hopf}}^2$ has not been computed for $Y^+(\widehat{\mathfrak{gl}}(1|1))$. Conjecture 12.6 accordingly remains open.

Conjecture 12.8 (Refined Z_{top} as $Y(\widehat{\mathfrak{gl}}(1|1))$ super-character at level 1). **[Conjectural]** (asserted as verified in the Wave-3 inscription; primary-literature status downgraded to conjecture here.) *Under $t = e^{i\varepsilon_1}$, $q = e^{i\varepsilon_2}$, the refined IKV formula $Z_{\mathrm{top}}^{\mathrm{ref}}(\mathbf{Y}; t, q, Q) = \prod_{i,j \geq 1} (1 - Q t^{i-1/2} q^{j-1/2})$ equals, coefficient-by-coefficient, the Weyl–Kac super-character of the level-one vacuum module of $Y_{\varepsilon_1, \varepsilon_2}(\widehat{\mathfrak{gl}}(1|1))$.*

Remark 12.9 (Coefficient-by-coefficient check at Q^0, Q^1). **[Primary-lit partial]** Expanding the IKV product in Q :

- Q^0 -coefficient: 1, matching the super-Yangian vacuum vector.
- Q^1 -coefficient: $-\sum_{i,j \geq 1} t^{i-1/2} q^{j-1/2} = -\frac{t^{1/2} q^{1/2}}{(1-t)(1-q)}$, which is the refined Hilbert series of $\mathbb{C}_{t,q}^2$ weighted by $-(tq)^{1/2}$. The super-Yangian side is the graded multiplicity of the simple-root affine generator $e_{\alpha_1}^{(1)}(z)$ on the level-1 vacuum; this has character $\frac{(tq)^{1/2}}{(1-t)(1-q)}$ under the $\mathbb{Z}^2$ -grading (Gaiotto–Rapčák [arXiv:1703.00982](#) §3.2 with the sign from fermionic generator). The two agree up to the overall $-$ sign.
- $Q^{1/2}$ -coefficient: *absent* in the IKV product (the expansion is in integer powers of Q); a “ $Q^{1/2}$ ” coefficient match would require identifying Q in IKV with Q^2 in some super-character convention. The correct coefficient-by-coefficient statement is in $Q^0, Q^1, Q^2, \dots$ (integer powers), with the fractional factor $(tq)^{1/2}$ appearing *inside* each integer- Q^n coefficient via the half-shift $t^{i-1/2} q^{j-1/2}$, rather than as a “ $Q^{1/2}$ -term” on the IKV side or as a “ $q^{1/2}$ -coefficient” in a q -expansion.
- Q^2 -coefficient: expected to match the graded multiplicity of the quadratic span $e_{\alpha_1}^{(1)} e_{\alpha_1}^{(1)}(z, w)$ on the vacuum, modulo fermionic Serre relations; explicit check is outstanding.

The full all- Q^n match is *conjectural*; only Q^0, Q^1 have a direct first-principles verification in the synthesis corpus. The fractional factor $(tq)^{1/2}$ appears inside every integer- Q^n coefficient via the half-shift $(i - 1/2, j - 1/2)$; there is no fractional power of Q in the expansion.

13. κ_{ch} RECONCILIATION (MANUSCRIPT-INTERNAL)

Remark 13.1 (Three manuscript numbers near the conifold: subscript disambiguation, not one invariant with three readings). **[Proposition]** (*four- κ discipline per CLAUDE.md; the Polyakov attack-heal sequence below picks one chain-level convention and derives the apparent three-valuedness from three distinct mathematical objects.*) The Vol III manuscript currently carries three numerically distinct values in the κ -neighbourhood of the conifold:

- $\kappa_{\text{ch}}(\mathbf{Y}) = +1$ in `chapters/examples/toric_cy3_coha.tex` Cor. cor:conifold-shadow (canonical toric chapter; identified there as $\text{DT}_{(1,0)} = 1$, the single compact curve-class contribution, $\kappa_{\text{ch,GV}}$).
- A value 0 appearing in `chapters/examples/derived_categories_cy.tex:853` which is $\kappa_{\text{cat}} := \chi(\mathcal{O}_{\mathbf{Y}}) = 0$ for the non-compact total space (Künneth/retraction to $\mathbb{P}^1$), *not* a second value of κ_{ch} .
- A value -1 appearing in `chapters/theory/e1_chiral_algebras.tex:2454` which is a $\kappa^!$ -type (Verdier-dual) invariant or a bulk BV anomaly, not κ_{ch} itself.

Under the CLAUDE.md four- κ discipline ($\kappa_{\text{ch}}, \kappa_{\text{cat}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}}$, *never conflated*), there is no inconsistency: the three numbers live under three different subscripts and record three different mathematical objects. The synthesis’s former phrasing “three inequivalent values of $\kappa_{\text{ch}}(\text{conifold})$ ” was itself an AP-CY113 bare- κ violation; the true rectification is subscripted disambiguation at every manuscript occurrence, not a “complementary projections” narrative that would posit a single κ decomposing into three pieces.

The “ $\kappa + \kappa^! \in \{0, 8, 13, 250/3, 98/3\}$ on the canonical five-archetype $\mathbf{G}/\mathbf{L}/\mathbf{C}/\mathbf{M}/\mathbf{B}$ ceiling” in Theorem C (shared five-theorem core) is a *G-archetype ceiling statement* about landmark Calabi–Yau data with specified $\chi_{\text{top}}/24$ -regulated centre-curving, *not* a universal identity $\kappa_{\text{ch}}(X) + \kappa_{\text{ch}}^!(X) = 0$ on every toric CY_3 . Reading off $\kappa_{\text{ch,bulk}} = -1$ from $\kappa_{\text{ch,GV}} = +1$ via “ $\kappa + \kappa^! = 0$ ” is *not* a theorem; it is an ansatz that would require the conifold to sit at the $\mathbf{G}$ -row of the Theorem C ceiling, which it does (class $\mathbf{G}$ with shadow depth $r_{\text{max}} = 2$, Cor. cor:conifold-shadow) — but the $\mathbf{G}$ -row entry of the ceiling is $K^\kappa = 0$ for toric bases (`toric_cy3_coha.tex` line 251: $K^{\kappa_{\text{ch}}}(\text{conifold}) = 0$), not $K^\kappa = 8$. The 8-value is the $\mathbf{B}$ -row Mukai-enhanced K3 Heisenberg witness on $K3 \times E$, not the $\mathbf{G}$ -row conifold entry.

14. TRANSFER TO TORIC CY_3 , OBSTRUCTIONS FOR COMPACT CY_3

Proposition 14.1 (Transfer to toric CY_3). **[Proposition]** *The five-route construction of $\text{CoHA}(\mathbf{Y})$ extends to any local toric Calabi–Yau threefold X without compact 4-cycles, with the following adjustments:*

- (1) *The toric atlas replaces the two-chart cover with a multi-chart cover indexed by the toric fan; each chart remains $\mathbb{C}^3$ with a Jordan-triple W_{chart} .*
- (2) *The Klebanov–Witten quiver is replaced by the appropriate toric quiver with potential (e.g., the Beilinson quiver for local $\mathbb{P}^2$; the $\mathbb{C}^3/\mathbb{Z}_n$ McKay quiver for local A_{n-1}).*
- (3) *The bond factors $\varphi_{a \Rightarrow b}$ are read off the toric fan via Galakhov–Li–Yamazaki [arXiv:2106.01230](#).*
- (4) *Davison–Meinhardt integrality + PBW gives the BPS Lie algebra.*
- (5) *The Drinfeld double is the appropriate affine (super-)Yangian of the root system determined by the quiver.*

This covers local $\mathbb{P}^2$, $\mathbb{C}^3/\mathbb{Z}_n$ for small n , banana threefolds, and other non-compact toric CY_3 without compact 4-cycles.

Theorem 14.2 (Sharp obstruction on compact CY_3 ; 2.5-count reduction). **[Proposition]** *The five-route construction does not transfer to compact Calabi–Yau threefolds. Four named obstructions — (O1) toric fan-completeness, (O2) $BCOV$ α_{BCOV} , (O3) Aut^0 rigidity, (O4) finite quiver with potential — reduce logically to 2.5 effective obstructions via $(O1) \Leftrightarrow (O3)$ and $(O1) \Rightarrow (O4)$, with (O2) the single independent BV-cohomological obstacle:*

$$(11) \quad (O1 \Leftrightarrow O3) \Rightarrow O4, \quad O2 \text{ logically independent.}$$

The effective count is $1 (O1 \equiv O3) + 1 (O2) + 0.5 (O4 \text{ downstream}) = 2.5$, not 4.

(O1) Failure of toric fan-completeness on a compact ambient. A toric atlas on a smooth threefold X consists of a complete rational polyhedral fan Δ in $N_{\mathbb{R}} \cong \mathbb{R}^3$ whose maximal cones $\sigma \in \Delta(3)$ are each generated by 3 primitive ray vectors (v_1, v_2, v_3) with $\det(v_1, v_2, v_3) = \pm 1$ (smoothness), subject to the Calabi–Yau slice condition: existence of a piecewise-linear support function $\psi : N_{\mathbb{R}} \rightarrow \mathbb{R}$ with $\psi(v_i) = 1$ for every ray v_i (Aspinwall–Greene–Morrison [hep-th/9309097](#) §3; equivalent to $c_1(X) = 0$ via the vanishing sum of ray vectors in each maximal cone). The simultaneous-slice condition $\psi(v_i) = 1$ forces every ray to lie on a common affine hyperplane $\{\psi = 1\} \subset N_{\mathbb{R}}$, so Δ is a cone over a 2-dimensional lattice polytope and has no compact 3-dimensional interior. The Demazure–Fulton criterion (Introduction to Toric Varieties, §3.4) yields: no smooth compact toric CY_3 exists; the only complete CY -sliced toric threefolds are finite quotients of $\mathbb{C}^3$. Compact CY_3 (the quintic, $K3 \times E$, bihypersurfaces) admit no chart-wise $(\mathbb{C}^3, W_{\text{chart}})$ cover, so the Schiffmann–Vasserot per-chart $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks have no geometric location on X ; the conifold’s Čech route of Section 5 is unavailable.

(O2) $BCOV$ 1-loop anomaly: independent BV cohomological obstacle. For a holomorphic Chern–Simons factorisation-algebra quantisation on a CY_3 X , Costello–Li 2016 [arXiv:1606.00365](#) Prop. 5.2 identify the unique 1-loop anomaly class

$$\alpha_{BCOV}(X) = \frac{\chi(X)}{24} [\Omega_X]^{0,1} \in H^{0,1}(X, \text{Sym}^{\leq 2} T_X^*),$$

sourced by the BV 1-loop counter-term Feynman graph weighted by Ω_X and the Todd specialisation $\chi(X)/24$. On $\mathbf{Y}$ the class vanishes by retraction $\mathbf{Y} \simeq \mathbb{P}^1$ and $H^{0,1}(\mathbb{P}^1) = 0$ (cf. Remark 11.5). On a compact CY_3 with $\chi(X) \neq 0$ and $[\Omega_X]^{0,1}$ non-vanishing in Dolbeault cohomology, $\alpha_{BCOV}(X) \neq 0$: no factorisation algebra of hCS observables exists at 1-loop. This obstruction sources from BV cohomology, not from automorphism geometry; it is logically disjoint from (O1).

(O3) Aut^0 -rigidity, equivalent to (O1). The Bogomolov decomposition theorem (Bogomolov 1974; Beauville 1983 JDG 18:755 §2) for a compact Kähler manifold X with $c_1(X) = 0$ gives a finite étale cover

$$\widetilde{X} = T^{2m} \times \prod_i S_i \times \prod_j Y_j,$$

with T^{2m} a complex torus, S_i irreducible hyperkähler, and Y_j strict Calabi–Yau ($h^{1,0}(Y_j) = 0$, $H^0(Y_j, \Omega_{Y_j}^2) = 0$). For a compact CY_3 of strict type (quintic, bihypersurfaces), $h^{1,0}(X) = 0$ and $h^{2,0}(X) = 0$ force $\text{Aut}^0(X)$ to act trivially on the Hodge decomposition; Matsumura 1963 Proc. Japan Acad. 39:181 Thm. 1 then concludes

$$\text{Aut}^0(X) = \{1\} \quad (\text{compact } CY_3 \text{ of strict type}).$$

For $X = K3 \times E$ of product type (stratum (ii) in Remark 14.5), $\text{Aut}^0(X) = E$ acts via translations on the elliptic factor; there is no $(\mathbb{C}^\times)^3$ torus action lifting the chart-wise conifold T^2 -equivariance. In either case, the conifold’s T^2 -fixed pyramid-partition localisation (Szendrői [arXiv:0705.3419](#) Thm. 2.6) has no analogue.

The equivalence $(O1) \Leftrightarrow (O3)$: a complete toric fan Δ for X generates a dense torus $(\mathbb{C}^\times)^3 \subset \text{Aut}^0(X)$, hence $\text{Aut}^0(X) \supsetneq \{1\}$ on a positive-dimensional stratum; conversely, a positive-dimensional $\text{Aut}^0(X)$ on a strict compact CY_3 is a complex torus by Bogomolov–Matsumura (the only connected

Lie groups preserving a CY_3 structure with $h^{1,0} = 0$ are abelian), and a complex torus acting faithfully with a dense orbit on a normal variety forces a toric structure (Sumihiro 1974 J. Math. Kyoto Univ. 14:1 Thm. 1). The two failures — absence of a toric fan, absence of a non-trivial connected automorphism group — are two facets of the same Bogomolov–Matsumura rigidity.

(O4) Finite quiver with potential, downstream from (O1). Van den Bergh’s NCCR construction (arXiv:math/0211064 Def. 4.1, Prop. 3.2.10) requires a tilting bundle $T \in D^b(\text{Coh } X)$ with $\text{End}_X(T)$ a finite-rank algebra over the Noetherian coordinate ring $R = \mathcal{O}_X(X)$. On a local toric CY_3 without compact 4-cycles, R is a finitely-generated $\mathbb{C}$ -algebra and the Bocklandt regularity criterion (arXiv:math/0603558 Thm. 3.1: $\text{NCCR} \Leftrightarrow$ the superpotential quiver is Calabi–Yau-3 in the Ginzburg sense and every arrow appears in W) promotes $\text{End}_X(T)$ to a quiver-with-potential $J(Q, W)$ with finitely many vertices and finitely many arrows, hence finite-dimensional per-vertex CoHA stalks $Y^+(\widehat{\mathfrak{gl}}_1)$. On a compact CY_3 , Hartogs on the projective base gives $R = \mathcal{O}_X(X) = \mathbb{C}$, so no finite-rank R -algebra tilting-endomorphism presentation can exist; $D^b(\text{Coh } X)$ is equivalent to no (Q, W) -Jacobi category. This is the contrapositive of (O1): no toric atlas $\Rightarrow$ no finite chart-wise quiver-with-potential $\Rightarrow$ no chart-wise finite-arrow CoHA. Hence (O1) $\Rightarrow$ (O4), and the automorphic-form replacement (Borcherds products, Gritsenko–Nikulin Δ_5 , Igusa Φ_{10}) enters via the non-chart-wise $(\infty, 1)$ -factorisation-homology route of Remark 14.4.

The five-route chart-wise programme terminates at the toric boundary for the single independent BV-cohomological reason (O2), the single independent geometric reason (O1 $\equiv$ O3), and the downstream representation-theoretic corollary (O4). Any compact CY_3 without a $K3 \times E$ -type product fibration admits no chart-wise $Y^+(\widehat{\mathfrak{gl}}_1)$ gluing; the replacement on $K3 \times E$ (Remark 14.6) is non-chart-wise Davison integrality with Maulik–Okounkov stable-envelope wall-crossing.

Remark 14.3 (Scope of the 2.5-count). Theorem 14.2 replaces any “four independent obstructions” phrasing with the 2.5-count derivation (11): (O1 $\equiv$ O3) is a single Bogomolov–Matsumura rigidity stated twice (toric-side and Aut^0 -side); (O4) is the downstream finite-quiver corollary via Van den Bergh tilting; (O2) is the logically independent BCOV 1-loop obstacle. The equivalence (O1) $\Leftrightarrow$ (O3) is *not* a coincidence of two independent cohomological tests; it is the twofold reading of one theorem (Bogomolov 1974 + Matsumura 1963 + Sumihiro 1974) — once as “no complete CY-sliced fan”, once as “ $\text{Aut}^0 = \{1\}$ on strict CY_3 ”. The BCOV class $\alpha_{\text{BCOV}} = (\chi/24)[\Omega_X]^{0,1}$ is independent because the Todd specialisation $\chi/24$ depends only on topological Euler data and $[\Omega_X]^{0,1}$ is a Dolbeault-cohomological representative unrelated to the fan structure.

Remark 14.4 (What does transfer). Three ingredients transfer to any CY_3 category:

- The Davison–Meinhardt critical CoHA + integrality PBW applies to any CY_3 category with a potential;
- Kontsevich–Soibelman motivic wall-crossing applies generally;
- The two-stage factorisation $\Phi_3 = \text{Sp}_{\Sigma_2, C}^{\text{ch}} \circ \Phi_3^{FA}$ is geometry-independent in principle.

What fails is the *explicit chart-wise formulas*: the BPS Lie algebra exists on any CY_3 , but its presentation for compact X is not chart-wise.

Remark 14.5 (The four equivariance strata). The four equivariance strata (CLAUDE.md):

- Toric* T^d (conifold, local $\mathbb{P}^2$, $\mathbb{C}^3$): chart-wise shuffle-algebra presentation, the setting of this paper;
- Reduced* $\mathbb{C}^\times \times \text{Aut}(X)$ ($K3$, $K3 \times E$, abelian surface): automorphism-enriched presentation, still equivariant but no toric atlas;
- Orbifold inertia* $I(X/G)$ (Mathieu M_{24} , McKay $\Gamma \subset SU(d)$): equivariant cohomology enriched by the inertia stack;
- Lattice-polarised period domain* (Borcherds lifts, Gritsenko Δ_5 , Igusa Φ_{10}): automorphic-form presentation with no chart-wise shuffle analogue.

The conifold lives in stratum (i); compact CY_3 typically live in strata (iii) or (iv); $K3$ and $K3 \times E$ straddle strata (ii)–(iv) depending on the Aut/inertia/Borcherds-lift chosen.

Remark 14.6 (Replacement machinery on compact CY_3). For compact CY_3 the CoHA is built via Maulik–Nekrasov–Okounkov–Pandharipande categorified machinery combined with Kontsevich–Soibelman motivic DT on the non-toric side. The $K3 \times E$ case (treated in Vol III Part IV of this programme) exemplifies the replacement: instead of two-chart $\mathbb{C}^3$ pieces, one applies Davison integrality directly to $D^b(\text{Coh}(K3 \times E))$, producing a Hall–Drinfeld double with imaginary rank-23 Cartan (the rank-24 Mukai lattice $\tilde{H}(K3, \mathbb{Z}) \simeq \Pi_{4,20} = H^0 \oplus H^2 \oplus H^4$ of dimension $1 + 22 + 1 = 24$, modulo the one-dimensional null direction spanned by the $\delta = [H^0] - [H^4]$ imaginary class corresponding to the E -fibre point/hyperplane exchange; see Chapter ?? and `chapters/examples/toric_cy3_coha.tex` line 181 for the sign $(2, 1)$ Cartan of $\mathfrak{g}_{\Delta_5}$), Humbert-surface monodromy of order 8, and Δ_5 -associator data — all non-toric invariants invisible to the chart-wise machinery. Chart-wise $Y^+(\widehat{\mathfrak{gl}}_1)$ stalks appear only at the 24 smooth- $\mathbb{C}^3$ points of a nodal- $K3 \times E$ degeneration, matching the 24 I_1 -fibres of the elliptic fibration on $K3$.

15. OPEN FRONTIERS

Four primary frontiers F_1 – F_4 govern the conifold programme; four secondary items F_5 – F_8 record auxiliary loci where the construction of §§3–14 admits sharpening. Each item carries a claim-status marker and an *explicit falsifier*: a first-principles computation whose outcome would disprove the item. Falsifiability is the Popperian standard demanded by Beilinson’s dictum; a conjecture that cannot be killed has no content. The items below each name the experiment that would kill them.

- F_1 . *Algebraic super-rigidity of $Y^+(\widehat{\mathfrak{gl}}(1|1))$.* **[Conjectural]** Conjecture 12.6: the Hopf-superalgebra cohomology $\text{HH}_{\text{Hopf}}^2(Y^+(\widehat{\mathfrak{gl}}(1|1)); Y^+(\widehat{\mathfrak{gl}}(1|1)))$ is one-dimensional, spanned by the class of the $\varepsilon_1 \varepsilon_2$ -deformation. *Falsifier.* Exhibit a 2-cocycle $\omega \in Z_{\text{Hopf}}^2$ on $Y^+(\widehat{\mathfrak{gl}}(1|1))$ linearly independent of $[\varepsilon_1 \varepsilon_2]$ modulo the $(\varepsilon_1 + \varepsilon_2)$ -gauge equivalence of Remark 12.7; equivalently, construct a formal deformation of the super-shuffle product (3) whose first-order class is not spanned by the IKV Schwinger kernel of Remark 12.5. Primary obstruction to the positive proof: Ginzburg–Schedler super-formality `arXiv:0807.0174` terminates at E_2 , removing the Kontsevich–Tamarkin E_3 -input that powered unrefined Schiffmann–Vasserot `arXiv:1202.2756`.
- F_2 . *Refined Z_{top} as super-character at level 1.* **[Conjectural]** Conjecture 12.8: $Z_{\text{top}}^{\text{ref}}(\mathbf{Y}; t, q, Q) = \chi_{Y_{\varepsilon_1, \varepsilon_2}(\widehat{\mathfrak{gl}}(1|1)), \text{lev } 1}^{\text{Weyl–Kac}}$ as formal power series in Q , with $t = e^{i\varepsilon_1}$, $q = e^{i\varepsilon_2}$. Coefficients Q^0, Q^1 verified directly (Remark 12.9); $Q^{\geq 2}$ open. *Falsifier.* Compute the Q^2 -coefficient of the IKV product in closed form and compare to the graded multiplicity $\text{ch}_{q,t}(e_{\alpha_1}^{(1)}(z) e_{\alpha_1}^{(1)}(w) \cdot |\text{vac}\rangle)$ in $Y_{\varepsilon_1, \varepsilon_2}(\widehat{\mathfrak{gl}}(1|1))$ modulo fermionic quadratic Serre; any mismatch not absorbed into the $(tq)^{1/2}$ half-shift normalisation of Remark 12.9 falsifies F_2 .
- F_3 . *Super-Yangian Hopf pairing asymmetric correction.* **[Conjectural]** Conjecture 9.10: the subleading coefficient on the odd diagonal is $c_{11} = -1$ in the Nazarov convention (equivalently $c_{11} = +1$ in the Gow–Peng antipode-in-coproduct convention), with $c_{00} = 0$ on the even diagonal. Leading binomial kernel verified via three independent routes (Maulik–Okounkov stable-envelope residue; Soibelman–Gelfand refined Ext-pairing $\mathbf{E}_h$; Tsymbaliuk shuffle `arXiv:1404.5240` Prop. 4.3). *Falsifier.* Perform the Berezinian-residue computation $\text{coeff}_{u-m-1, v-n-1}[\log \text{Ber}_q T(u) \cdot k^{\text{off}}(u-v)]$ with $k^{\text{off}}(u) = (u + \varepsilon_1)(u + \varepsilon_2)/[u(u + \varepsilon_1 + \varepsilon_2)]$ in the Nazarov 1991 / Gow `arXiv:math/0411166` / Peng 2011 framework. A value $|c_{11}| \neq 1$ falsifies the parity-isotropy-source argument; a non-zero c_{00} falsifies the even-self-Serre input on the Cartan diagonal.

- F₄. *M-theory parent and 5D hCS super all-orders*. **[Conjectural]** Conjecture 11.1 with Theorem 11.3 at 2-loop and higher: the 6D (2, 0) A_{K-1} theory on $\mathbb{R}^{1,3} \times T^2$ with K M5-branes wrapped on $\mathbb{P}^1 \subset \mathbf{Y}$, normal-bundle twist $\mathcal{O}(-1) \oplus \mathcal{O}(-1)$, Ω -deformation on transverse Taub–NUT, and Costello–Gwilliam twist on the CY_3 target quantises after T^2 -reduction to $Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ to all orders in $\hbar$. *Falsifier*. Compute the 2-loop Costello–Paquette BV-BRST anomaly cocycle on the super-abelian (1|1)-stack; a non-trivial cohomology class at $O(\hbar^2)$ obstructs the factorisation-algebra quantisation and falsifies the all-orders claim at $K = 1$. For $K \geq 3$: the cubic-Casimir anomaly $A_{\text{anom}} = d^{abc}d^{abc}/(2\pi)^3$ obstructs non-abelian extension; $K = 2$ is special because $d^{abc} = 0$ on $\mathfrak{sl}_2$. Explicit Costello–Paquette super-extension covering the $(-1, -1)$ -twisted normal bundle is absent from current literature.

Secondary frontiers, auxiliary to the master identification:

- F₅. *Hopf pairing non-degeneracy beyond weight (2, 2)*. **[Primary-lit partial]** Shapovalov determinants computed up to weight (2, 2) (Wave 2 Agent 1) with generic non-degeneracy verified and two Kazhdan–Lusztig-positive chambers separated by the conifold BPS wall $\hbar = 1$. *Falsifier*. Compute $\det \text{Shap}_{(3,3)}$ as a polynomial in $\hbar$ and exhibit a zero off the Kac denominator locus $\prod_{r,s}(\hbar - \hbar_{r,s})$; such a zero falsifies the conjectured higher-weight generic non-degeneracy and the KL-positivity persistence across the BPS wall.
- F₆. *Chamber-character match at higher weight*. **[Primary-lit partial]** Coefficient-by-coefficient agreement across R1–R5 verified at weights $m + n \leq 4$; higher weights open. *Falsifier*. Any pair R_i, R_j producing distinct structure constants at a weight (m, n) with $m + n \geq 5$, after chamber normalisation (Theorem 10.2) and the native-basis intertwiner $\iota^{R_i R_j}$ of Theorem 16.1(i), falsifies the five-route match.
- F₇. *Compact CY_3 replacement machinery*. **[Open]** Theorem 14.2 establishes four obstructions to direct transfer; the *positive* side — an explicit automorphic-form replacement on compact CY_3 — is open. Gritsenko–Nikulin `alg-geom/9504006` and Borchers `alg-geom/9609022` supply partial data on $K3 \times E$ (Vol III Part IV). *Falsifier*. Construct a non-toric compact CY_3 admitting a finite quiver with potential (Q, W) with Ginzburg dga vanishing $\text{HH}^{<0}(J) = 0$ whose CoHA admits a chart-wise presentation equivariant under a continuous torus of rank ≥ 2 ; any such example falsifies the simultaneous necessity of all four obstructions (O1)–(O4).
- F₈. χ_{BLLPRvR} *identification at $\mathcal{N} = 1$* . **[Conjectural]** The 4D/2D chiral-algebra functor of Beem–Lemos–Liendo–Peelaers–Rastelli–van Rees `arXiv:1312.5344` is defined for $\mathcal{N} \geq 2$ 4D SCFTs; $T[\mathbf{Y}]$ at the resolved conifold is $\mathcal{N} = 1$. The claim $Y^+(\widehat{\mathfrak{gl}}(1|1)) = \chi_{\text{BLLPRvR}}(T[\mathbf{Y}])$ requires an $\mathcal{N} = 1$ -compatible extension, e.g. via R -symmetry enhancement at the IR superconformal point, or via a partial-topological twist recovering $\mathcal{N} = 2$ bookkeeping on a sub-Coulomb locus. *Falsifier*. Exhibit the IR superconformal fixed point of $T[\mathbf{Y}]$ and show its $U(1)_R$ -current does not enhance to $SU(2)_R$; equivalently, prove that no $\mathcal{N} = 1 \rightarrow \mathcal{N} = 2$ enhancement exists in the deep IR of $T[\mathbf{Y}]$, closing the applicability of χ_{BLLPRvR} .

Each F_i is falsifiable by a concrete computation of bounded combinatorial complexity; none is a metaphysical conjecture. The four primary items F_1 – F_4 share the obstruction — absence of E_3 -super-formality and of Costello–Paquette super-extensions to CY_3 with $(-1, -1)$ -twisted normal bundle — consistent with these being four facets of a single open problem: the chain-level quantisation of the super-BPS E_3 -centre.

16. MASTER THEOREM

Theorem 16.1 (Master identification). **[Proved elsewhere]** (at the super-algebra level, primary source Li–Yamazaki `arXiv:2003.08909` §8.3.6; at the ungraded-shadow level, primary source MMNS `arXiv:1107.5017` combined with Davison–Meinhardt `arXiv:1601.02479` integrality.) *Let $\mathbf{Y} = \text{Tot}(\mathcal{O}_{\mathbb{P}^1}(-1) \oplus \mathcal{O}_{\mathbb{P}^1}(-1))$ be the resolved conifold and $(Q_{\text{con}}, W_{\text{con}})$ the Klebanov–Witten*

two-vertex quiver with quartic cyclic potential. Write $Y^+(\widehat{\mathfrak{gl}}(1|1))$ for the positive half of the affine super-Yangian of $\widehat{\mathfrak{gl}}(1|1)$ and $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ for its conifold specialisation (Li–Yamazaki [arXiv:2003.08909](#) §8.3.6, bond factors (3)). Then

$$\boxed{\text{CoHA}(\mathbf{Y}) \xrightarrow{\cong} Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}}$$

is an isomorphism of $\mathbb{Z}_2$ -graded associative bialgebras in super-vector spaces over $\mathbb{C}((\hbar))$, with structure maps $(\mu, \iota, \Delta, \epsilon)$ identified as follows:

- (a) Multiplication μ . The critical CoHA convolution product on $\text{CoHA}(\mathbf{Y})$ coincides with the super-shuffle star product on $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ with bond factors (3) (Li–Yamazaki eq. (8.125)).
- (b) Unit ι . The identity class $[\text{pt}] \in \text{CoHA}(\mathbf{Y})_{(0,0)}$ maps to the vacuum $1 \in Y^+$.
- (c) Coproduct Δ . The Kontsevich–Soibelman localised integration map ([arXiv:1006.2706](#) §6.3) equals the Drinfeld-new coproduct $\Delta(e^{(a)}(z)) = e^{(a)}(z) \otimes 1 + \phi^{+, (a)}(z) \otimes e^{(a)}(z)$ in $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$, where $\phi^{+, (a)}(z) = \sum_{k \geq 0} \phi_k^{+, (a)} z^{-k-1}$ is the Drinfeld-new positive Cartan current on vertex $a \in \{0, 1\}$.
- (d) Counit ϵ . $\epsilon|_{\text{CoHA}_{(0,0)}} = \text{id}_{\mathbb{C}}$, $\epsilon|_{\text{CoHA}_{(m,n) \neq (0,0)}} = 0$.

An antipode is **not** asserted on $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ itself: the positive half is a bialgebra but not a Hopf algebra. The antipode lives on the Drinfeld double $D(Y^+)$ alone; its scope is recorded in item (ii) below.

- (i) Five presentations produce isomorphic bialgebras. The five routes R1–R5 (Sections 4, 5, 6, 7, 8) each produce a bialgebra; any two are related by a native-basis intertwiner $\iota^{\mathbf{R}_i \mathbf{R}_j}$ that is an $\hbar$ -linear bialgebra isomorphism on every weight-graded piece $(\text{CoHA})_{(m,n)}$ with $m+n \leq 4$, agreeing at the leading $(1,1)$ -weight imaginary-root generator with coefficient $(\varepsilon_1 \varepsilon_2)^{-1}$ in each native basis. Agreement at $m+n \geq 5$ is F_6 (Section 15). Routes R1, R4, R5 resolve the central Cartan current $K(z) = h^{(0)}(z) + h^{(1)}(z)$ via residue extraction at the coincidence locus $z^{(0)} = z^{(1)}$; routes R2, R3 resolve the Chevalley Cartan $H(z) = h^{(0)}(z) - h^{(1)}(z)$ and project K to zero under the chart-wise $Y^+(\widehat{\mathfrak{gl}}_1)$ -stalk stencil (Remark 3.3). The intertwiner $\iota^{\mathbf{R}_i \mathbf{R}_j}$ absorbs both the native-basis change and the $H \leftrightarrow K$ reassignment on the imaginary-root line.
- (ii) Drinfeld double and Hopf status. The double $D(Y^+) = Y(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$ carries an antipode; it is a strict $\mathbb{Z}_2$ -graded Hopf algebra at rational, trigonometric, and toroidal spectral type (Etingof–Kazhdan cohomology $H_{\text{EK}}^3(\widehat{\mathfrak{gl}}(1|1); \widehat{\mathfrak{gl}}(1|1)^{\otimes 3}) = 0$, Theorem 9.12), and a quasi-Hopf algebra with dynamical associator Φ_{ell} (Felder–Jimbo–Konno) at elliptic spectral type.
- (iii) BPS Lie algebra. $\mathfrak{g}_{\text{BPS}}^{\text{con}}$ has Klebanov–Witten affine root structure: real roots $n\delta \pm \alpha$ of multiplicity 1, imaginary roots $n\delta$ of super-dimension 2, spanned by $\{H_n, K_n\}$ with H the Chevalley Cartan and K the central Cartan (Proposition 3.5).
- (iv) Chamber structure. The stability space is a $W(\widehat{\mathfrak{sl}}_2)$ -torsor of shifted quiver Yangian representations of a chamber-independent algebra; the Joyce–Song–Kontsevich–Soibelman pairing is literally equal across chambers, not merely projectively equivariant (Theorems 10.1, 10.2).
- (v) Ungraded shadow is a different algebra (AP-CY305). The supertrace projection $\text{str}: \widehat{\mathfrak{gl}}(1|1) \rightarrow \widehat{\mathfrak{sl}}_2$ on the even subalgebra annihilates the central current K and retains only the Chevalley H (Remark 3.3). It induces a surjection $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}} \twoheadrightarrow Y^+(\widehat{\mathfrak{sl}}_2)^{\text{con}}$ of bialgebras whose kernel is the two-sided ideal generated by K . The target $Y^+(\widehat{\mathfrak{sl}}_2)^{\text{con}}$ is the MMNS motivic DT algebra [arXiv:1107.5017](#) and is a distinct algebra from $Y^+(\widehat{\mathfrak{gl}}(1|1))^{\text{con}}$: the imaginary-root line of the super source is two-dimensional ($H + K$), while the shadow’s imaginary-root line is one-dimensional (H only). Both are theorems at their respective scopes — the super form is primary for the categorified CoHA (Li–Yamazaki [arXiv:2003.08909](#) §8.3.6), the shadow is primary for motivic/numerical invariants (MMNS + Davison–Meinhardt integrality [arXiv:1601.02479](#)) — and the relation between them is surjection, not isomorphism.

Transfer to arbitrary local toric CY_3 without compact 4-cycles is Proposition 14.1; the four-way obstruction to direct transfer to compact CY_3 is Theorem 14.2; the replacement automorphic-form machinery on $K3 \times E$, quintic, and Gritsenko–Nikulin–Borcherds-lift compact CY_3 is Remark 14.6 and frontier F_7 (Section 15).

Remark 16.2 (Scope summary: what is proved, what is not). The isomorphism $\text{CoHA}(\mathbf{Y}) \cong Y^+(\widehat{\mathfrak{gl}(1|1)})^{\text{con}}$ of item (i) is established as bialgebras on weight-graded pieces with $m + n \leq 4$ via explicit structure-constant match across R1–R5; extension to all weights is the conjectural content of frontier F_6 . The antipode claim of item (ii) is established on $D(Y^+)$, not on Y^+ itself. The AP-CY305 discipline of item (v) distinguishes the super bialgebra from its ungraded shadow as *two different algebras* (surjection of the super form onto the shadow), not as two presentations of one algebra. The refined $(\varepsilon_1, \varepsilon_2)$ -structure underlying Conjectures 12.6 and 12.8 is the open content of frontiers F_1, F_2 . Hopf-pairing non-degeneracy at weight $> (2, 2)$ and the $c_{11} = -1$ asymmetric correction are F_5, F_3 .

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